

Vertical and Horizontal Agency Costs: Evidence from Public and Private Firms

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Abstract

We investigate vertical (ownership-versus-control) and horizontal (majority-versus-minority) agency costs based on multiple accounting and executive compensation based measures, and a sample of over 42,000 public and private firms. We find that agency costs are higher in public firms relative to private firms. Among private firms, we show that agency costs are: (a) higher when firms are not managed by owners; (b) increase as firms move from simple to complicated ownership structures; (c) lower in firms with shared control of ownership; and (d) lower in firms where control is contestable. Tests that account for endogeneity also fully support our conclusions.

Keywords: Agency problems, Control, Horizontal agency costs, Management structure, Ownership structure, Private companies, Vertical agency costs.

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Vertical and Horizontal Agency Costs: Evidence from Public and Private Firms

Agency problems arise when managers or controlling shareholders have the ability to redirect or consume corporate resources in ways that benefit themselves but which are not in the best interests of the other owners, including minority owners.¹ Firms face two types of agency problems: “vertical” agency problems that exist between owners and managers arising from the separation of ownership and control (Jensen and Meckling, 1976), and “horizontal” agency problems between controlling (majority) shareholders and minority owners arising from their potentially conflicting interests (Shleifer and Vishny, 1986; Gilson and Gordon, 2003).² Tirole (2006) underscores two important manifestations of agency problems: (a) inefficient investment choices, which could include the redirection of resources for personal consumption; and (b) inefficient or insufficient effort being expended by managers. The loss in performance and the resulting costs that arise as a result of these and related inefficiencies are generally referred to as agency costs.³ The magnitude of any agency costs should arguably depend on potentially measurable factors such as inefficient asset utilization (arising from poor investments or, for instance, the appropriation of resources) by those who control the decision making process; excessive and unwanted production costs and perks (resulting in higher expenses); and insufficient effort exerted by management (resulting in lower revenues and earnings).

We analyze an extensive dataset on ownership and management, which includes details on

¹The extensive literature on agency issues stems from the seminal work of Berle and Means (1932), Jensen and Meckling (1976), Ross (1973), and includes, *inter-alia*: Shleifer and Vishny (1986), Holmstrom and Kaplan (2001), Holmstrom and Kaplan (2003), Becht, Bolton, and Röell (2003), Denis and McConnell (2003), Hermalin (2005), Gilson and Gordon (2003), Tirole (2006), and Djankov, La Porta, Lopez-de Silanes, and Shleifer (2008).

²The vertical agency problem emerges from the view of the firm as a nexus of contracts that are costly to enforce (Coase, 1937; Alchian and Demsetz, 1972; Ross, 1973; Jensen and Meckling, 1976; Fama and Jensen, 1983b,a). In contrast, the literature on horizontal agency problems has focused on the exploitation of minority shareholders by a controlling shareholder (Grossman and Hart, 1980; Stulz, 1988; Burkart, Gromb, and Panunzi, 1997, 1998; Gilson and Gordon, 2003; Dyck and Zingales, 2004; Laeven and Levine, 2008). See also the review by Roe (2005).

³Jensen and Meckling (1976) include in the definition of agency costs the costs expended by agents on bonding and by principals on monitoring. What we refer to as agency costs essentially are what Jensen and Meckling (1976) call the ‘residual loss’ due to the less than optimal use of resources that arises because of the agency problem.

over 42,000 firms (and over 109,000 firm-year observations): including both "public" firms, i.e., firms whose shares are publicly traded and consequently widely held; and "private" firms, i.e., firms that are not publicly traded, and hence with a limited-span and relatively closely held ownership structure.⁴ We use this dataset to study the relation between measures that represent agency costs and the structure of ownership and control in an organization. We empirically examine the relations between our agency cost proxies and the full spectrum of ownership and management arrangements ranging from the single-owner single-manager firm to the diffusely held corporation with many managers, as well as the relation to the structure of ownership *amongst* different owners.

Our study makes several contributions to the literature. First, we focus on the entire ownership and management spectrum: from single-owner-single-manager firms through to diffusely held publicly traded corporations. While providing important insights, studies focusing on the valuation effects of agency problems almost always concentrate on publicly-traded companies.⁵ These studies are not able to address agency costs associated with the vertical or horizontal agency problems over the complete ownership spectrum that also spans private companies; even though such private companies represent the overwhelming majority of corporate entities globally. We present such evidence.

We find that private firms exhibit lower agency-related costs when compared to public firms in our sample. For instance, our results suggest that on average, public firms are associated with lower asset turnover ratios (0.16 standard deviations lower), higher operating expenses to sales ratios (0.25 standard deviations higher), and lower earnings (0.25 standard deviations lower), compared to a matched sample of private firms.⁶ We also find that the level of executive com-

⁴Close corporations in the U.S. are examples of a "private" firm ownership structure.

⁵The literature is extensive. See the survey articles of Shleifer and Vishny (1997), Denis and McConnell (2003), Gillan (2006) as well as Morck, Shleifer, and Vishny (1988), McConnell and Servaes (1990), Himmelberg, Hubbard, and Palia (1999), Demsetz and Villalonga (2001), Coles, Lemmon, and Wang (2008). In related work Yermack (2006), Andrews, Linn, and Yi (2015) and Grinstein, Weinbaum, and Yehuda (2011) conclude perquisite consumption is at least in part due to vertical agency problems.

⁶These specific estimates are based on the results presented in models M4-M6 of Table 4.

compensation is higher, and the pay-for-performance sensitivity lower, among public firms compared to private firms (Hartzell and Starks, 2003). Further, among private firms, we find that owner-managed firms, defined by us as firms in which managers own more than 99% of the firm, exhibit evidence of lower agency-related costs when compared to private firms with other ownership and management structures, consistent with the theoretical predictions of Jensen and Meckling (1976). These results are robust to various control variables as well as statistical methods.

We also find that among private firms, agency costs increase as firms move from simple ownership structures, such as being owned by a single individual or a family, to more complicated ownership structures, such as being owned by multiple families or holding companies. This result is consistent with the hypothesis that, as the number of shareholders increases and the types of shareholders increase, the incentive for any individual shareholder to incur the costs of monitoring managers decreases, because the benefits associated with monitoring are proportional to the shareholder's ownership stake (Shleifer and Vishny, 1986; Holderness, 2009) - which in turn results in less monitoring than would arise in the case of fewer shareholders.

The second contribution of this paper is an investigation of the costs associated with the horizontal agency problem. While both vertical and horizontal agency problems exist in private firms, many argue that the horizontal agency problem, which manifests itself in the exploitation of minority shareholders by a controlling shareholder, is a major concern in private firms and close corporations (Gilson and Gordon, 2003; Burkart et al., 1997, 1998; Dyck and Zingales, 2004; Roe, 2005; Laeven and Levine, 2008). The exploitation of minority shareholder wealth can take several forms including higher compensation for majority shareholders, either directly or indirectly, through transfer pricing arrangements between related companies that favor the controlling shareholder; and the direct appropriation of corporate assets by the controlling shareholder, for instance through excess dividend payments. Corporate inefficiencies are a manifestation of such actions. In order to attract minority investors, the majority shareholder has an incentive to credibly convey that minority shareholders will not be exploited. Theory

suggests that the controlling shareholder should transfer some control to minority shareholders as a means of providing a credible promise that minority owners will not be exploited (Pagano and Roell, 1998; Bennedsen and Wolfenzon, 2000; Shleifer and Wolfenzon, 2002; Gomes and Novaes, 2005).

We find that firms with shared control, defined by us as firms where the largest shareholder owns less than 50% of the equity, have lower agency costs than firms in which the largest shareholder owns more than 50% of the equity, and thereby has enough power to extract private benefits from minority shareholders. Furthermore, we find that the presence of multiple large shareholders also results in lower agency costs. We document an inverse relation between the ownership stake of the second largest shareholder and agency costs, providing support for the view that minority expropriation is lower in companies where control is more contestable.⁷ These results hold for both firms that are owner-managed and those that are non-owner managed.

We also present results that arguably account for endogeneity and are consistent with our main findings. These tests allow us to disentangle the endogeneity issues that can potentially arise because of the possibility that ownership structure, management structure and agency costs may be jointly determined. In these tests we examine how different types of firms responded to a major exogenous macroeconomic shock. More specifically, we appraise how the firms in our sample responded to the 2008 financial crisis. The force of these tests is that while such a shock should increase financial stress on firms in general, we would not expect it to directly affect governance quality, ownership and management structures in the short-term around the shock as for instance concluded by Johnson, Boone, Breach, and Friedman (2000), Mitton (2002) and Lemmon and Lins (2003) in another context. We find that the financial crisis results in an increase in the agency costs of public firms relative to private firms; non-owner-managed firms relative to owner-managed firms; and firms with no contestability in control

⁷These results are consistent with the findings of Lehmann and Weigand (2000), Faccio, Lang, and Young (2001) and Maury and Pajuste (2005).

relative to firms in which control is contestable. These results are completely consistent with our results documented in the rest of the paper, given extant findings already reported in Mitton (2002) and Lemmon and Lins (2003) that such a shock increases the incentives for managers to expropriate owners (vertical agency costs) and for majority shareholders to expropriate minority shareholders (horizontal agency costs). Our results are also robust to a variety of additional specifications of critical variables and the use of different data samples.

Finally, we analyze the extent to which actual executive compensation deviates from what would otherwise be the normal level under the hypothesis that firms characterized by greater potential agency problems should be associated with greater agency costs which might manifest as compensation in excess of what would have been normal. We relate this deviation to ownership and management structure and find that the results also confirm our overall conclusions. Similarly, we find supporting results based upon models of the change in compensation.

The general absence of audited and verified information on close corporations or private firms including ownership structure is the primary reason for the lack of empirical evidence on agency costs measured across the entire ownership spectrum. A private firm owned and managed by a single individual clearly has zero agency costs. The zero agency cost firm serves as the baseline case when measuring the agency costs incurred by firms with different management and ownership structures. However, no publicly traded firm can be classified as a zero agency cost firm because, by definition, such firms are characterized by a separation of ownership and control. While it is relatively easy to obtain audited and verified financial information about public companies, such information is typically unavailable for private companies, and in particular for private U.S. companies.

However, the financial reporting regulations for U.K. companies are substantially equivalent for private and public firms. Three principal features of these regulations motivate our use of U.K. data, and provide assurance regarding its quality and relevance in the context of our study. First, the U.K. Companies Act requires all private and public companies to file annual financial statements that comply with the same accounting standards. Second, financial state-

ments filed by both U.K. private as well as public companies must be audited. Third, private and public companies are subject to the same tax laws. These standards therefore ensure that we have a set of comparably measured data for both public and private companies. In addition, and importantly, the data filed provides information on the ownership structures of the filing companies, both private and public. Our dataset includes firms with a wide array of ownership structures ranging from firms with a single owner-manager to firms with multiple owners and outside managers.

Our paper is close in spirit to the work of Ang, Cole, and Lin (2000). Ang et al. (2000) examine self-reported data on small businesses collected via telephone survey methods as part of the National Survey of Small Business Finances (NSSBF). Using a sample of 1,708 domestic U.S. companies for 1992, Ang et al. (2000) find that their proxy for agency costs is significantly higher when an outsider manages the firm and is inversely related to the manager's ownership share. Using data from the same survey, Nagar, Petroni, and Wolfenzon (2011) find that firms with shared ownership are associated with a larger return on assets. Our study differs from these two papers in several important ways. First, to our knowledge this is the first study to compare agency costs in public and private companies. Second, our conclusions are based on data of constituent firms that reliably follow a common set of duly audited reporting guidelines (unlike the NSSBF data used in earlier research which was voluntarily reported through telephone surveys, and consequently unaudited). The companies represented in our sample are all domiciled in the U.K. and must comply with government mandated reporting requirements. Third, our sample is truly comprehensive: our tests are based on a sample that includes over 42,000 unique private as well as public companies, and the data examined cover the multiyear period 2005-2010; and our robustness tests also include tests covering over 200,000 unique firms and half a million firm-year observations. The cross-section of companies we examine is diverse across both industries as well as company size. Fourth, the NSSBF data used by the above-mentioned papers contains limited measures of ownership. In contrast, our dataset contains detailed information on ownership, and because we have multiyear observations, we are able

to utilize lagged data on ownership and management as a precaution against joint determination of these variables and agency costs.⁸

In section 1 we discuss the nature of the agency costs associated with both vertical and horizontal agency problems, and outline testable research questions. Section 2 provides a description of the sample, and contains a description of how we measure agency costs. Section 3 describes the methodology and the control variables used in the study. Sections 4 and 5 discuss empirical results on vertical agency costs and horizontal agency costs respectively. Section 6 presents a discussion of the effects of an exogenous shock in the form of the 2008 financial crisis on agency costs. Section 7 presents results using abnormal executive compensation as a proxy for agency costs. Section 8 presents additional robustness tests. Section 9 presents a summary of the paper and our conclusions.

1 Research Questions

1.1 Vertical Agency Costs

Jensen and Meckling (1976) argue that when an owner-manager reduces her equity stake below 100%, incentives increase for the owner-manager to consume or waste corporate resources for personal benefit because she does not bear the full cost of such excesses. These "vertical" agency costs increase as the owner-manager's ownership decreases. The simplest ownership structure is one in which a single individual owns and manages the firm. Such firms represent the zero-agency cost base case. Such a single owner-manager could have subordinate managers reporting to her, potentially leading to costs arising from misalignment of interests and inadequate monitoring. However, while the sole owner bears 100 percent of any agency costs, she is also fully incentivized to use her exclusive right to hire or fire subordinate managers to

⁸The NSSBF data provides four measures of ownerships structure: a) the ownership share of the primary owner; b) an indicator for firms where a single family controls more than 50 percent of the firm's shares; c) the number of non-manager shareholders and d) an indicator for firms managed by a shareholder rather than an outsider.

effectively monitor and discipline subordinate managers since she receives 100 percent of the benefits from such monitoring. The monitoring is expected to weaken when there are multiple owners, and the lead owner-manager holds less than 100% equity ownership. As we move from a single owner-manager setting to structures where firms are owned by multiple shareholders, we expect the magnitude of vertical agency costs to increase. As the number of shareholders increases, the incentive for any shareholder to incur all of the cost of monitoring the managers decreases because the benefits associated with monitoring are limited by the shareholder's proportional ownership stake. The reduced incentive to monitor in turn may lead to higher agency costs. A potential remedy to this problem is to have simpler and more concentrated ownership structures. Shareholders with a larger stake will have a greater incentive to monitor managers and see that agency costs are kept low. Agency costs are therefore predicted to increase as firms move from the single-owner structure to more complex ownership structures. Likewise, the magnitude of vertical agency costs are predicted to increase as the proportion of shareholders managing the firm decreases.⁹

1.2 Horizontal Agency Costs

A fundamental feature of close corporation and private firm ownership structures is that shareholders are typically few in number, are knowledgeable about firm operations, and are involved in management. In particular, when a controlling shareholder is present, that person generally takes an active interest in running the company by choosing the management and directly holding an executive position. While concentrated ownership helps mitigate the vertical agency problem, it is possible that a controlling shareholder will extract private benefits of control by forcing decisions which expropriate minority shareholder wealth (Grossman and Hart, 1980; Dyck and Zingales, 2004; Gilson and Gordon, 2003; Roe, 2005).

⁹Empirical evidence on the monitoring role of large shareholders of publicly traded companies is extensive. Please see Shleifer and Vishny (1986), Wruck (1989), Franks, Mayer, and Renneboog (2001) and references cited in footnote 2. A recent example is Cronqvist and Fahlenbrach (2009) who find blockholder fixed effects in policy choices involving investment, financial, and executive compensation as well as in firm performance measures.

Pagano and Roell (1998) suggest that by monitoring the controlling shareholder, other large shareholders play an important role in reducing horizontal agency costs. Gomes and Novaes (2005) speculate that the presence of a large number of blockholders improves firm governance in closed corporations because disagreement among shareholders prevents them from expropriating minority shareholders. In a model developed by Bennedsen and Wolfenzon (2000), no individual shareholder has sufficient votes to control the firm, and consequently must form a coalition of shareholders to achieve control. Coalition formation minimizes the chance of expropriation since no individual shareholder is able to take any actions without the consent of the other coalition members. A result is that fewer choices expropriating minority shareholders are implemented, and firm performance is better relative to the single controlling shareholder case. The main shareholder surrenders some control to minority shareholders in order to improve overall firm performance. The prediction is that shared control of firms helps decrease the magnitude of horizontal agency costs.¹⁰

Pagano and Roell (1998) specify conditions under which multiple large shareholders will cross-monitor each other, reducing expropriation and improving firm performance. In their model, expropriation of minority shareholders by a controlling shareholder is likely to be less severe when the ownership stake of non-controlling shareholders is more concentrated. The intuition behind this conclusion is that large non-controlling shareholders are more effective in monitoring the controlling shareholder. In a related analysis, Bloch and Hege (2003) conclude that minority expropriation will be lower in firms where control is more contestable, that is in firms where the difference in the stakes of the controlling shareholders and that of minor-

¹⁰Faccio et al. (2001) examine the effect of multiple large shareholders on dividends. They find that the presence of large shareholders dampens expropriation in Europe (due to monitoring), but exacerbates it in Asia (due to collusion). Lehmann and Weigand (2000) show that the existence of a second large owner is positively associated with the profitability of listed German firms. Maury and Pajuste (2005) investigate a sample of listed Finnish firms and conclude that a more equal distribution of votes among large blockholders has a positive effect on firm value. Gutiérrez Urtiaga and Tribó (2004) examine Spanish firms and find that firms whose characteristics make them more vulnerable to minority expropriation tend to have controlling groups with ownership stakes that are far removed from a 50% threshold. In a related study, Berkman, Cole, and Fu (2011) examine the links between political connections and minority shareholder protection in Chinese listed firms.

ity shareholders is smaller. An empirical implication of these theories is that the magnitude of horizontal agency costs decreases as contestability increases.¹¹

1.3 Questions Addressed

In the light of sub-sections 1.1 and 1.2 above, we address the following specific research questions:

- a Are agency costs higher in public firms when compared to private firms?
- b Are agency costs higher in non-owner-managed firms when compared to owner-managed firms?
- c Are agency costs higher in firms with a larger number of shareholders, or with more complex ownership structures, when compared to firms with fewer shareholders or less complex ownership structures?
- d Does shared control of ownership help reduce agency costs?
- e Does contestability of control help reduce agency costs?

2 Sample Construction and Measures of Agency Costs

2.1 Sample Construction

Our sample consists of financial and ownership data on public and private companies registered in the United Kingdom. The U.K. Companies Act of 1967 (and subsequent revisions to the act) requires all private and public companies that cross a designated size threshold to file

¹¹Lehmann and Weigand (2000), in a study of publicly traded German companies, report that the presence of a second large shareholder enhances profitability. In a related paper, Volpin (2002) analyzes listed Italian companies and finds that firms where control is contestable have higher valuations.

annual financial statements that comply with the same accounting standards. All accounts submitted to the Companies House must be audited and certified by an independent accountant. Companies designated as “small” and “medium” sized can claim exemptions from either or both the audit requirement and the accountant’s report requirement, and are also allowed to file abridged financial statements.¹² Small companies can be exempt from both audit and accountant’s report requirements, whereas medium companies can be exempt only from auditing requirements but do need an accountant’s report.¹³

We obtain company data from the FAME database produced by *Bureau Van Dijk*. Each FAME update contains the latest available ownership and management structure data and up to 10 years of financial statement data for all public and private companies registered in the U.K. and Ireland. We use updates dated October 2005, October 2006, November 2007, October 2008, October 2009 and October 2010 to extract data.¹⁴ We identify all firms classified as private limited, public quoted, public quoted AIM, public quoted OFEX, and public not quoted.¹⁵ We require that the firms have minimum assets of \$5,000 and sales of \$1,000 to avoid selecting non-operating firms. These initial sample criteria result in 1,332,576 firm-year observations. After excluding firms with missing SIC codes and firms with missing information on the type of accounts filed, we obtain a sample of 1,303,956 firm-year observations with financial information available. We then perform the following steps to filter the sample. Please refer to Appendix A

¹²The definition of a “small” or “medium” company has changed over time. For accounting periods ending after 30 January 2004, a company is considered small (medium) if one of the following requirements are met: i.) annual revenues are less than 5.6 (22.8) million, ii) balance sheet total is 2.8 (11.4) million or less and iii.) number of employees is less than 50 (250).

¹³ For more details please refer to <http://www.companieshouse.gov.uk/about/gbhtml/gba3.shtml#three>.

¹⁴The update dated October 2005, for instance, contains financial information for fiscal years 1995-2004 and the latest ownership data depending on firms’ survival status.

¹⁵Firms belonging to the public quoted category are those that are listed on the London Stock Exchange (LSE). Firms belonging to the public quoted AIM group are those that are listed on the Alternative Investment Market (AIM). OFEX is an independent public market specializing in smaller companies who do not wish to incur the expense of listing with AIM. Firms with different ownership structures might differ in their accounting practices such as recognition and timing of revenues and costs because of tax considerations. All U.K. firms, both public and private, are subject to the same tax laws (Ball and Shivakumar (2005); and Bell and Jenkinson (2002)) so tax code driven ownership structures should not be present in our sample.

for a more detailed description of the sample construction.

1. We restrict our sample to fiscal years 2002 and later, losing 4,252 observations.
2. We require that firms have lagged ownership data available. We are able to find lagged ownership data for 869,113 observations. Tables A.2 and A.3 in the appendix provide details.
3. We retain firms that have both ownership and financial information available. Further, we exclude firms in the financials and utilities industries, and firms with missing assets, sales and operating expenses. This leaves a sample of 500,087 observations (appendix table A.4).
4. Lastly, we exclude firms that are exempt from the auditing requirements of the Companies House. This results in our losing 390,533 more observations. This selection criterion leads to exclusion of extremely small private firms and ensures all the firms in our sample face the same accounting and reporting regulations.¹⁶

Hence, the results reported in the tables are based on a final sample of 109,534 observations spanning fiscal years 2002 to 2010. There are a total of 6,156 public firm-year observations corresponding to 1,761 unique public firms and 103,378 private firm-year observations corresponding to 40,698 unique private firms in our sample. presents the yearly distribution of the full sample and the different sub-samples. There are relatively fewer observations prior to fiscal year 2004 and after fiscal year 2009.¹⁷

We further classify our sample of private firms into sub-samples based on ownership and management structures. More specifically, we classify these firms into the following groups: firms owned by a single individual, firms owned by a single family, firms owned by multiple

¹⁶A potential concern with this criterion is possible sample selection bias from imposing the size threshold implied by the auditing requirements of the Companies House. We address this issue in Section 8.

¹⁷There are only 33, 280 and 1,060 firm-year observations from fiscal years 2002, 2003 and 2010 respectively. Hence, we merge observations from these years into fiscal years 2004 and 2009 for reporting purposes.

families and firms owned by a combination of individuals and private industrial entities such as holding companies. Further, within each group, we identify a set of firms that are owner-managed – defined by us as firms in which managers own more than 99% of the firm. Of particular interest are the firms that are owned and managed by a single individual. There are 3,644 such firms and 7,427 firm-year observations covering a wide-array of industries. This group of firms can be classified as the zero-agency cost firms defined by Jensen and Meckling (1976).

Panel B of Table 1 presents information on the industries to which the sample firms belong. Classification of firms into industry groups is done according to the one-digit U.K SIC code. There are significant variations in how private and public firms are distributed among industry sectors. While only 13% of private firms belong to the manufacturing sector, nearly 30% of public firms belong to this sector. On the other hand, we observe that the proportion of firms belonging to the construction and real estate sectors is higher in private firms. In sum, these differences highlight the importance of controlling for industry effects in the analyses to follow.

Panel A of Table 2 presents company-level summary statistics for the public and private firms in our sample. To mitigate the effect of outliers, we winsorize the top 1 percent and bottom 1 percent of all financial variables. Private firms are typically smaller and younger than public firms. The mean (median) size of private firms in our sample is \$32.5 million (\$4.1 million) whereas the corresponding value for public firms is \$906.9 million (\$44.9 million) respectively. Private firms in our sample are, on average, more levered (average debt-to-asset ratio of 41.6% compared to 23.7% for public firms), consistent with the findings reported by Brav (2009). While private firms are smaller and more levered than public firms, they are associated with higher turnover and profitability ratios, as measured by asset turnover (AT) and earnings before interest taxes and depreciation scaled by assets (EBITD) respectively.¹⁸ We also report on “Quiscore”, a measure of the probability of company failure (widely used commercially in

¹⁸The negative average values of EBITD/Assets for our sample of public companies are likely driven by the bottom quartile of public companies. We verify the accuracy of our data by examining data on U.K. public companies available in the Compustat Global and Datastream databases and note that the average EBITD/Assets calculated from these databases are negative during the sample period as well.

the U.K.), computed using a proprietary model and based on variables conceptually similar to those employed in the computation of the Altman Z-score.¹⁹ The distribution of Quiscores across ownership classifications indicates the index values for the sample firms tend to fall in the normal or better regions. With respect to ownership and management structures, we were able to account for 99.1% of the ownership stakes of private firms on average and approximately 72% for public firms. Average managerial ownership is 57% for private firms and 17% for public firms where data are available. Owner managed firms account for nearly 18% of private firms.

Panel B of Table 2 presents company-level summary statistics for private firms further classified into four different ownership structures as defined above. Firms owned by a single individual are on average smaller and younger than firms owned by other groups. With respect to management structures, a majority of single owner firms (77%) are owner-managed.²⁰ Interestingly, a significant proportion of all private firms have a bank loan, and the average bank debt to total assets ratio decreases monotonically as we go from firms owned by a single individual to firms owned by multiple individuals and holding companies.

2.2 Measures of Agency Costs

Important manifestations of agency problems are inefficient investment choices, and inefficient or insufficient effort being expended by managers (Tirole, 2006). We can accordingly proxy

¹⁹Quiscore is produced by CRIF Decision Solutions Ltd. It is a measure of the likelihood of company failure in the year following the date of calculation. In determining a Quiscore value for a company, a number of separate calculations are performed using various combinations of financial characteristics including turnover (revenue), pre-tax profit, working capital, intangibles, cash and bank deposits, creditors, bank loans and overdrafts, current assets, current liabilities, net assets, fixed assets, share capital, reserves and shareholders' funds. The Altman Z-score requires the ratio of the market value of equity to the book value of equity for its computation and so cannot be directly computed for private companies.

²⁰The vast majority of family owned firms are also owner-managed. The reason why only 6.5% of family owned firms appear to be owner-managed is because a large proportion of firms owned by a single family have two individuals listed as shareholders, but only one of those shareholders is listed as a director. Since our definition of an owner-managed firm requires that managers own at least 99% of the firm (to closely replicate Jensen and Meckling's (1976) definition of a zero agency cost firm), a majority of firms owned by a single family are (presumably incorrectly) classified as not owner-managed. To the extent that owner-managed firms are associated with low agency costs, we note that this misclassification, if anything, biases our results against rejecting the null hypothesis of no significant differences between low agency cost firms and high agency cost firms.

agency costs by measures based on: (a) relative inefficiencies in asset utilization (arising from poor investments), (b) production cost inefficiencies (arising from excessive and unwanted production costs and perks), and (c) inefficiencies in overall value generation resulting in relatively lower earnings (because of insufficient effort exerted by management).

Since the overwhelming majority of our sample companies are private companies, our proxies for agency costs cannot be based on market-based data.²¹ Our agency cost proxies are accordingly based on accounting and executive compensation data. Specifically, our first measure is the efficiency of asset utilization using the asset turnover ratio ("AT"), which reflects how management uses the assets under control for revenue generation (Ang et al., 2000; Singh and Davidson III, 2003). Our second measure is production cost efficiency using operating expenses divided by sales ("OPEXP") (Ang et al., 2000; Nagar et al., 2011). Our third measure is the ratio of earnings before interest taxes and depreciation to total assets ("EBITD"), and captures aggregate efficiencies in overall value creation. Finally, in the spirit of (Hartzell and Starks, 2003), we also use a non-accounting based measure as a proxy for agency costs, and examine how the level of executive compensation and the pay-per-performance sensitivity of executive compensation vary among firms with different ownership structures relative to what normal values for these variables would equal conditional on company characteristics. Because no single measure can capture the loss of performance arising from agency problems comprehensively, we use all these measures in our empirical analyses.²²

²¹Studies examining publicly traded companies use market-based measures like a pseudo-Tobin's Q measure or return on equity to infer agency costs. Market-based measures are clearly unavailable for (non-traded) private companies.

²²We report results using executive compensation measures in a separate section as this data are available for fewer firms.

3 Research Design and Control Variables

3.1 Methods

We use broadly two approaches in our examination of the questions posed earlier. The first is a matched sample analysis. We perform a matched sample analysis of firms that are likely to be associated with higher agency costs (treatment group) and those that are likely to be associated with lower agency costs (control group) based upon the theories articulated in section 1. For example, while comparing public and private companies, for each public firm (treatment group), we find a matching private firm (control group). Our matching procedure works as follows. For each observation in the treatment group, we first match it with observations in the control group belonging to the same fiscal year, two-digit SIC code and with total assets within 30% of the treatment observation. Among the matches, we identify an observation with closest QuiScore and total debt. The matching (with replacement) procedure helps mitigate potentially large differences in the size distribution between the two samples and any other differences that may occur due to industry affiliations and the likelihood of company failure. In a majority of tests, the number of firm-year observations in the treatment group far outnumbers the number of firm-year observations in the control group. We therefore randomly choose 5,000 observations from the treatment group and proceed with the matching criteria just described.

In order to examine how agency costs vary over time between a matched pair of firms in the treatment and control group, we also create a matched sample of firms using a balanced panel. A balanced panel is created by limiting the sample period to fiscal years 2005 to 2009 and including only those firms that are a.) not associated with changes in ownership structure during the period and b.) present in the sample every year between 2005 and 2009. We then create a matched sample using firms in fiscal year 2005 and follow these matched pairs throughout the sample period.

Our second approach is based on ordinary least squares regressions pooled across owner-

ship types. We standardize all variables except dummy variables in each sample at the industry (two-digit SIC code) and year level. All regression specifications include industry fixed effects and year fixed effects and standard errors are clustered at the year-industry level.²³

3.2 Control Variables

In all our regression specifications we control for firm size, firm age, bank borrowing, general leverage, default risk, and an indicator of accounting disclosure. Company size may be associated with the extent or lack of an agency problem. Williamson (1967, 1985) for instance suggests economies of scale and other related factors influence the size of the firm, but that decreasing returns to managerial efficiency, and span of control issues, may emerge in larger firms. Conversely, large firms may be those which have survived and grown due to operating efficiently. We control for the size of the firm using total assets. Older firms may be more efficient than younger firms and the fact that they have survived may suggest the agency costs for such firms are smaller. On the other hand older firms may reflect situations where investment opportunities have been largely exhausted and excess cash flow permits greater abuse of resources. We therefore also control for firm age. We calculate firm age as the number of years between the incorporation date and financial statement date.

In the absence of access to public equity markets, private firms rely on debt provided by owners and external institutions such as banks as their primary source of financing. For instance, Berger and Udell (2002) and Cole, Wolken, and Woodburn (1996) note that financial institutions provide roughly 27% of the dollar amount of small business credit in the US with 16% additionally coming from trade credit. The role of monitoring has long been recognized as an important ingredient of bank lending (Diamond, 1991; Tirole, 2006). We assume that a bank's monitoring incentives are directly proportional to the level of loans they make to a firm. We include bank debt scaled by total assets as an independent variable. We also include total

²³In separate analyses (not reported) we used propensity score based matching methods and regressions based on Fama and MacBeth (1973) methodology. Results were qualitatively similar.

liabilities scaled by total assets as a control. We control for the likelihood of company failure in the subsequent year by including dummy variables for four of the five bands (high risk, caution, normal and stable) into which the Quiscore of a company falls with the "secure" band being the excluded group. We include the number of subsidiaries and number of holding companies associated with a firm as control variables.²⁴

An important issue in studies focusing on ownership structure is whether company ownership structure and firm performance are endogenous relative to one another. Put simply, while ownership structure may affect performance, it is possible that performance may also be one of the determinants of ownership structure. We address this issue formally by examining how different types of firms respond to the exogenous shock of the 2008 financial crisis, that arguably increases financial stresses without concomitantly changing the quality of governance, or the ownership and management structures.

Moreover, we also examine whether ownership structure for our sample is sticky, that is, we examine whether ownership structure tends to vary over time for our sample companies.²⁵ Constant ownership levels would be a sufficient condition to motivate its use as an independent variable in a performance-type regression (Smith and Watts, 1992). We examine our sample for changes in ownership structures and find that less than 2% of the private firms in the sample were associated with an ownership change from any one year in the sample to the next. We exclude these firms and repeat our analysis. The results are qualitatively similar to those we report in this paper, and are available on request. Even though we find that ownership structure does not tend to vary over time, we use lagged ownership and management classification data in our analysis to minimize any possibility of contemporaneous jointly determined effects.²⁶

²⁴We redo our analysis by excluding private firms with subsidiaries. The results do not vary significantly from those presented.

²⁵While it might be possible for an owner to increase her equity investment, she will be constrained by her own personal resources (Berger and Udell, 2002; Cole et al., 1996). In most private company settings, especially those cases involving a single owner, most of the owner's wealth is likely to already be tied up in the company she owns. As a result of this resource constraint and the fact that such ownership interests are not easily transferable, investors in private firms and close corporations cannot easily adjust their ownership positions as conditions change.

²⁶Ownership data are on average lagged by 345 days relative to the date of filing financial statements.

4 Results - Vertical Agency Costs

4.1 Public and Private Firms

We first compare agency costs in public and private companies. Agency theory predicts that as the fractional ownership stake of a manager decreases, and as the number shareholders increases, agency costs increase. Thus, if agency costs are material, these costs should be higher among public firms when compared to private firms. Given the large difference in the size distribution between public and private firms (see Table 2), we create a matched sample of public and private firms using the matching criteria described in section 4.1. Given that we match on size, industry, year, likelihood of company failure and total debt, we feel comfortable in attributing the differences in performance measures between public and private companies to costs arising because of agency problems. We create an alternate matched sample wherein matching is done in fiscal year 2005 and the matched pairs are then followed throughout the sample period.

Table 3, Panel A presents the results of the matched sample analysis using all private and public firms. We were able to find matches for 5,163 of the 6,156 public firm observations. The median size of the public firms in the matched sample is \$29.5 million whereas the median size of the private firms in the matched sample is \$27.4 million (not reported). We conduct one sample t-tests and Wilcoxon sign rank tests to test the null hypothesis that the mean (median) of paired differences for each performance measure equals zero. The results unequivocally indicate that, for each of the three agency cost proxies, public firms are associated with considerably and significantly higher agency cost proxies compared to private firms. For instance, the average value of the ratio of operating expenses to sales is about 23 percentage points higher in public firms when compared to the matched sample of private firms. Similarly, the asset turnover ratio for public firms is about 21 percentage points lower relative to private firms. Results using the balanced panel are presented in Panel B and are qualitatively similar. To mitigate any concerns that the results are driven by any particular year, we repeat this analysis at a yearly level and present the results in Panels C through H of Table 3. Our finding that public firms are

associated with higher agency costs is robust to this analysis. We next estimate the following regression specification:

$$\begin{aligned} \text{Agency Cost Proxy}_{it} = & \alpha + \beta_P * \text{Public}_{it-1} + \gamma_{Size} * \text{Assets}_{it} + \gamma_{Firm} * \text{Firm Controls}_{it} + \\ & + \delta_{Industry} * \text{Industry Dummies} + \delta_t * \text{Year Dummies} + \varepsilon_{it}, \end{aligned} \quad (1)$$

where "Public" is a dummy variable which equals 1 for public companies and 0 otherwise. "Firm Controls" include firm age, bank loan, total liabilities, number of subsidiaries, a dummy variable "group" that equals 1 if group accounts are filed and 0 otherwise, a dummy variable "dividend dummy" that equals 1 if firms pays a dividend and 0 otherwise, and dummy variables "stable band", "normal band", "caution band" and "high risk band" that equal 1 if the QuiScore of the company falls between 60-80, 40-60, 20-40 and below 20 respectively and 0 otherwise.

Table 4, Panel A presents results using the matched sample of public and private firms. Recall all variables except for dummy variables in each sample are standardized at the two-digit industry and year level. The coefficient estimate on "Public" in model M2 indicates that public companies are associated with a higher operating expense to sales ratio of 0.27 standard deviations compared to private companies. To the extent the higher expenses are the result of inefficiencies arising due to the presence of agency problems, we can conclude that agency costs are higher in public firms. Coefficient estimates on "Firm Controls" are not presented for the sake of brevity and are available upon request.

Models M4-M6 in Panel A present results with managerial ownership and an interaction between managerial ownership and "Public" as additional explanatory variables. The coefficient estimate on the interaction variable captures the incremental effect of manager ownership in public companies relative to manager ownership in private companies. While manager ownership helps reduce agency costs as indicated by the positive (negative) and significant coefficient estimates on the manager ownership variable in models M4 and M6 (M5), it is important to note that public companies still exhibit higher agency costs compared to private companies. Models

M1-M3 in Panel B present results using a matched sample created from a balanced panel of public and private firms where as models M4-M6 present results using the full sample of public and private firms. The results are qualitatively similar.

4.2 Owner-Managed and Non-Owner-Managed Firms

We now focus on how agency costs vary among private firms. The results presented in Tables 5 and 6 emphasize the difference in agency costs between owner-managed and non-owner-managed firms. Panel A of Table 5 presents the results of a matched sample analysis. The control sample is a group of firms that are owner-managed and the treatment sample consists of firms that are not owner managed. Given that non-owner-managed firms outnumber firms that are owner-managed, we randomly choose 5,000 non-owner-managed firm-year observations and match those with owner-managed firm-year observations. Focusing on the operating expenses to sales measure, we note that the mean and median of the paired differences are both positive and statistically significant indicating that this measure is higher for non-owner-managed firms compared to owner-managed firms. Panel B presents results using a matched sample based on a balanced panel of owner-managed and non-owner-managed firms. Results are qualitatively similar. Panel C presents results of a matched sample analysis of firms owned by a single individual. Owner-managed firms in this group are, in many respects, examples of zero-agency cost firms. We find that owner-managed firms outperform non-owner-managed firms in this sub-group as well and the differences are statistically significant at the 1% level for all three measures of agency costs. We next estimate the following regression specification:

$$\begin{aligned} Agency\ Cost\ Proxy_{it} = & \alpha + \beta_{OM} * Owner - Managed_{it-1} + \gamma_{Size} * Assets_{it} + \gamma_{Firm} * Firm\ Controls_{it} \\ & + \delta_{Industry} * Industry\ Dummies + \delta_t * Year\ Dummies + \varepsilon_{it}, \end{aligned} \quad (2)$$

where "Owner-Managed" is a dummy variable that equals 1 if managers own more than 99% of the firm and 0 otherwise. Panel A of Table 6 presents the results using a matched sample. We expect the coefficient estimate on "Owner-Managed" to be positive in models M1 and M3

and to be negative in model M2. Results are consistent with this expectation. For example, the coefficient estimate on "Owner-Managed" is -0.13 and significant at the 1% level in model M2 indicating that on average, owner-managed firms are associated with operating expenses that are 0.13 standard deviations lower than non-owner-managed firms, accounting for the controls. The results are consistent with the hypothesis that agency costs are larger in firms not managed by owners. These results, along with those presented by Ang et al. (2000), indicate that the hypothesis is supported for firms domiciled in both the U.K. and the U.S.

We provide additional sets of tests to further examine the robustness of our conclusions. We replace the dummy variable "Owner-Managed" with a continuous measure of manager ownership. The results are presented in models M4-M6 of Panel A and suggest that agency costs decrease with an increase in the fractional ownership of managers. Panel B of Table 6 presents results using a matched sample created from a balanced panel (models M1-M3) and the full sample of private firms (models M4-M6). The results are consistent with the above discussion.

A salient feature of this study is that it provides an opportunity to look at various ownership structures among private firms. As mentioned earlier, we divide our private firm sample into those owned by a single owner, one family, multiple families and individuals and companies respectively. We next examine whether agency costs resulting from separation of ownership and control are relevant in each of the ownership structures. We find that higher manager ownership helps reduce agency costs in each ownership structure, except that the difference is not significant in some specifications for firms owned by a single owner or a single family. The results also indicate that the magnitude of the effect increases as we move from simple to more complex ownership structures. These results are not presented for the sake of brevity and are available on request. Overall, we conclude from the results presented in Tables 5 and 6 that agency costs are significantly greater for non-owner managed firms and that this conclusion is robust.

4.3 Ownership Structure and Agency Costs

While the results presented in Tables 5 and 6 lead to the general conclusion that agency costs are larger in non-owner managed firms, one specific hypothesis also suggests that as the number of shareholders increases the incentive for any single shareholder to incur all of the cost of monitoring the managers decreases because the benefits associated with monitoring are limited by the shareholder's proportional ownership stake, which is less than 100%. The reduced incentive to monitor in turn may lead to higher agency costs. We test this prediction using our sample of private firms.

We restrict our sample to private firms where at least 90% of the ownership stake is accounted for, and those with non-missing values of AT, OPEXP and EBITD. We then sort this sample on number of shareholders and compare observations in the top quartile (with highest number of shareholders) with those in the first two quartiles (with fewer shareholders). Accordingly, our treatment sample in this case consists of firms with more than five shareholders and the control sample consists of firms with less than or equal to two shareholders. We are able to find matches for 4,854 observations of the 5,000 randomly chosen firm-years with more than five shareholders. Results are presented in Panel A of Table 7 and support the prediction of a direct relationship between the number of shareholders and the magnitude of agency costs. Firms with more shareholders are associated with a larger operating expense to sales ratio and lower asset turnover and earnings ratios. The paired differences are statistically significant at the 1% level for two of the three measures while insignificant for the asset turnover measure. Panel B presents results using a matched sample created from a balanced panel of firms. Results are qualitatively similar to those presented in Panel A.

We next examine whether manager ownership, proxying for alignment of interest between owners and managers, drives the positive association between number of shareholders and the magnitude of agency costs. In order to test this, we repeat our matched sample analysis using two subsamples. The first subsample consists of firms with above median manager ownership and the second subsample consists of firms with below median manager ownership. The results presented in Panels C and D of Table 7 indicate our main findings are robust to these

specifications. We next run the following regression specification:

$$\begin{aligned}
 \text{Agency Cost Proxy}_{it} = & \alpha + \beta_{NS} * \text{Number of Shareholders}_{it-1} + \gamma_{MO} * \text{Managerial Ownership}_{it-1} \\
 & + \gamma_{Size} * \text{Assets}_{it} + \gamma_{Firm} * \text{Firm Controls}_{it} + \delta_{Industry} * \text{Industry Dummies} \\
 & + \delta_t * \text{Year Dummies} + \varepsilon_{it},
 \end{aligned} \tag{3}$$

Table 8 presents results of regression analysis. Models M1-M3 present results using the matched sample. Coefficient estimates on “Number of Shareholders” have the correct signs in all three models and are statistically significant at the 1% level in two of the three models. For instance, model M2 presents results with the ratio of operating expenses to sales as the dependent variable. The coefficient estimate on the number of shareholders indicates that, on average, a one standard deviation increase in the number of shareholders increases the operating expense to sales ratio by 0.082 standard deviations. Models M4-M6 present results using the full sample of private firms and results support the positive association between complexity of ownership structures and agency costs.²⁷ Models M1-M3 (M4-M6) in Panel B of Table 8 present results using a matched sample of firms with above median (below median) levels of manager ownership. The results presented suggest that even after controlling for manager ownership, firms with complex ownership structures are associated with higher agency costs, although the results are slightly weak for the subsample of firms with high manager ownership. The coefficient estimates on “Number of Shareholders” have the correct signs in all specifications, and are statistically significant in four of the six models. Overall, these results provide support for the hypothesis that the magnitude of agency costs increase as firms move from simple to complex (generally more diffuse) ownership structures.

²⁷We also use ownership structures to proxy for the reduced incentive to monitor resulting from a larger number of shareholders and complex ownership structures. We define two dummy variables “Firms Owned by Multiple Families” and “Firms Owned by Individuals and Companies” that equal 1 if the firm is owned by multiple families and a combination of individuals and companies respectively and 0 otherwise. The excluded group includes firms owned by a single individual or a single family. The coefficient estimate on both the dummy variables is positive but is statistically significant on the latter. Results are not presented for the sake of brevity and are available on request.

4.4 Summary

The results presented in this section indicate: 1) companies managed by a non-owner exhibit larger agency costs, 2) companies which exhibit more diffuse ownership exhibit larger agency costs, 3) the marginal effect on vertical agency costs of the presence of a non-owner manager, conditional on the level of ownership, is positive. These results are broadly consistent with the basic propositions of agency theory in Jensen and Meckling (1976) and other papers.

5 Results - Horizontal Agency Costs

Horizontal agency costs arising from the expropriation of minority shareholders by a majority shareholder are likely to be more severe in close corporations (Roe, 2005; Gillan, 2006). This is likely to be true regardless of whether or not those firms are owner-managed. In this section, we focus exclusively on private firms with more than two shareholders in our examination of this issue.

5.1 Ownership Concentration

We begin by testing whether greater shared control is associated with smaller horizontal agency costs. Unlike public firms where owners can exert decision control with a relatively low ownership stake (Gillan, 2006), private firms can be controlled only by owning at least 50% of the firm.²⁸ Private firms are owned by few shareholders and therefore, it would be relatively easy for shareholders owning a stake greater than 50% to block the decisions of any owner who owns less than a 50% stake. Dominant large shareholders have more complete control and, thus, have a greater opportunity to engage in decisions that allow them to extract private benefits at the expense of minority shareholders: i.e., to consume resources whose value exceeds the personal loss they incur as a result.

²⁸For evidence consistent with this view see Morck et al. (1988) and LaPorta, Lopez-de Silanes, and Shleifer (1999). Dyck and Zingales (2004) also use a 50% cutoff threshold.

Results of a matched sample analysis are presented in Panel A of Table 9. We classify firms into two groups. The treatment sample includes firm-years where the largest shareholder owns more than 50% (absence of shared control) and the control sample includes firm-years where the largest shareholder owns less than 50% (presence of shared control) of the firm. The mean and median values of the paired differences for two of the three efficiency variables are all statistically significant indicating that firms with shared control are associated with higher performance measures reflecting lower agency costs. For instance, firms without shared control are associated with asset turnover ratios that are 5.2 percentage points lower on average compared to firms with shared control. Panel B presents results of a matched sample created from a balanced panel. Results are qualitatively similar to those presented in Panel A. We next estimate the following regression specification:

$$\begin{aligned}
 \text{Agency Cost Proxy}_{it} = & \alpha + \beta_{LC} * \text{Low Concentration}_{it-1} + \gamma_{MO} * \text{Managerial Ownership}_{it-1} + \\
 & \gamma_{Size} * \text{Assets}_{it} + \gamma_{Firm} * \text{Firm Controls}_{it} + \delta_{Industry} * \text{Industry Dummies} \\
 & + \delta_t * \text{Year Dummies} + \varepsilon_{it},
 \end{aligned} \tag{4}$$

where "Low Concentration" is a dummy variable that equals 1 if the ownership stake of the largest shareholder is less than 50% and 0 otherwise. When included in a model with an intercept, the coefficient estimate on "Low Concentration" measures the change in agency costs of the diffusely held firm relative to high concentration firms. Models M1-M3 in Panel A of Table 10 present results using a matched sample and models M4-M6 present results using the full sample of private firms with more than two shareholders. The results indicate that low concentration firms are associated with lower agency costs relative to high concentration firms. The coefficient estimate on "Low Concentration" is positive and statistically significant in all models with AT and EBITD as the dependent variables and is negative (but insignificant) in models with OPEXP as the dependent variable. Overall the results support the view that shared control of ownership helps reduce agency costs in private firms or closed corporations.

5.2 Shareholder Monitoring and Control Contestability

As discussed earlier, a complimentary hypothesis is that the magnitude of horizontal agency costs decreases as control becomes more contestable. The agency costs associated with the horizontal agency problem are predicted to be smaller when a large, non-controlling, shareholder is present to monitor and contest decisions of the controlling shareholder. We test this hypothesis using two measures defined to capture the impact of the second largest shareholder on the level of agency costs. The first measure is the ownership stake of the second largest shareholder and the second is the difference between the stakes of the largest and the second largest shareholders (Difference).

We first present the results of a matched sample analysis. The treatment sample consists of firms with below median values of second largest shareholder's ownership stake (absence of contestability) and the control sample consists of firms with above median values for the second largest shareholder's ownership stake (presence of contestability). We randomly choose 5,000 firm-year observations from the treatment group and are able to find matches for 4,818 observations. Results of this analysis are presented in Panel A of Table 11. The results suggest that firms with a powerful second largest shareholder are associated with lower agency costs when compared to firms with a weak second largest shareholder. For instance, firms with a powerful second largest shareholder are associated with asset turnover ratios (operating expenses to sales ratios) that are 16 percentage points (5.7 percentage points) higher (lower) than firms with a weak second largest shareholder. Panel B of Table 11 present results of a matched sample created from a balanced panel and results are similar to those presented in Panel A. We next estimate the following regression specification:

$$\begin{aligned}
 \text{Agency Cost Proxy}_{it} = & \alpha + \beta_{MC} * \text{Measure of Contestability}_{it-1} + \gamma_{MO} * \text{Managerial Ownership}_{it-1} \\
 & + \gamma_{Size} * \text{Assets}_{it} + \gamma_{Firm} * \text{Firm Controls}_{it} + \delta_{Industry} * \text{Industry Dummies} \\
 & + \delta_t * \text{Year Dummies} + \varepsilon_{it},
 \end{aligned} \tag{5}$$

We use three measures to proxy for contestability. Table 12 presents the results. Models M1-M3 present results using a matched sample. The variable of interest in these models is a dummy variable ("Contestable") that equals 1 for firms associated with above median values of second largest shareholders ownership stake and 0 otherwise. We expect the coefficient estimate on this variable to be positive and significant when AT and EBITD are the dependent variables and negative and significant when OPEXP is the dependent variable. Results are consistent with our expectations. Next, we replace the dummy variable with a continuous measure of the ownership stake of the second largest shareholder and repeat our tests using the full sample of private firms. Results are presented in models M4-M6 of Table 12. These results are consistent with non-controlling shareholders acting as monitors, as well as acting to contest decisions which would be detrimental to minority shareholders; thus, making private benefit extraction more costly for the controlling shareholder.

Contestability may also be a function of the difference in the sizes of the respective holdings of the largest and next largest shareholders. If the share of the largest shareholder is much larger than all other owners, the ability of smaller owners to prevent expropriation may be limited. Put differently, horizontal agency costs may be positively related to the difference in the ownership stakes of the largest and the second largest shareholders. We test this prediction by including the difference in ownership stakes ("Difference") as an explanatory variable and present the estimation results in Models M7-M9 Table 12. Focusing on the results presented in model M8, we note that the estimated coefficient on "Difference" is positive and statistically significant. That is, the larger the difference between the stakes of the top two shareholders of the firm, the higher are agency costs.

Overall the results presented in this subsection are consistent with the model proposed by Pagano and Roell (1998). That is, minority shareholder expropriation is lower in companies where control is more contestable, companies where the difference between the stakes of the majority and minority shareholders is smaller.

6 Tests Using the Financial Crisis as an Exogenous Shock

In this section, we use the relatively exogenous shock of the 2008 financial crisis to help disentangle the endogeneity issues that can potentially arise because of the possibility that ownership structure, management structure, and agency costs may be jointly determined. Specifically, we examine whether differences in ownership and management structures can explain differences in our observed agency cost measures that arose in response to the 2008 financial crisis in a manner that is consistent with our hypotheses and our earlier results.

As has been argued and illustrated elsewhere (Mitton, 2002; Lemmon and Lins, 2003), a financial crisis corresponds to an unexpected negative shock to investment opportunities and managerial wealth without affecting ownership and management structures, at least in the short term. During times of prosperity when managerial wealth and income are high, the incentive for managers of high agency problem firms to engage in expropriative activities that increase agency costs is reduced. This incentive shifts in times of economic distress during which wealth and income receive a negative shock. In the latter state, the incentive to consume perquisites or engage in other activities that increase agency costs to the benefit of managers or large owners increases. Consistent with this expectation, Mitton (2002) empirically finds that stock returns *during* the east asian financial crisis are significantly lower for firms associated with greater agency problems compared to returns of firms associated with lower agency problems. In a related paper, Lemmon and Lins (2003) also find evidence consistent with this conclusion.

In view of the above, if our hypotheses and earlier results are correct, we should arguably find that the exogenous shock represented by the 2008 financial crisis should have resulted in a greater increase in agency costs for (a) public firms relative to private firms, (b) non-owner managed firms relative to owner managed firms, and (c) firms without contestability in control relative to firms with contestable control structures. We test this conjecture by employing the

following difference-in-difference analysis:

$$\begin{aligned} \text{Agency Cost Proxy}_{it} = & \alpha + \beta_{HA} * \text{High Agency Indicator}_{it-1} + \beta_{Crisis} * \text{Crisis}_t + \\ & \beta_{HAC} * (\text{High Agency Indicator}_{it-1} * \text{Crisis}_t) + \gamma_{Size} * \text{Assets}_{it} + \\ & \gamma_{Firm} * \text{Firm Controls}_{it} + \delta_{Industry} * \text{Industry Dummies} + \varepsilon_{it}, \quad (6) \end{aligned}$$

where “Crisis” is a dummy variable that equals 1 for fiscal year 2008 and 0 otherwise. We exclude fiscal years 2007 and 2009 from this analysis as the exact start and end dates of the crisis are unclear.²⁹ “High Agency Indicator” is a variable proxying for firms that we have hypothesized as having relatively high agency costs and varies based on the sample used. In tests comparing public and private firms, public firms are treated as the group hypothesized to have high agency costs. Then, “High Agency Indicator” is a dummy variable that equals 1 for public firms and 0 otherwise. Similarly, when we compare owner-managed and non-owner-managed firms, the non-owner-managed firms are the hypothesized high agency cost group; and when we test for different horizontal agency costs, the firms without contestability in control are hypothesized as high agency cost firms. The coefficient estimate on the interaction variable captures the average change in performance for high agency firms (relative to the comparison group) attributable to the financial crisis. Appendix B describes the other variables that appear in the model. As in previous regressions, all continuous variables are standardized at the year, industry level.

Models M1-M3 of Table 13 present results comparing private and public firms. Here public firms are hypothesized as the group associated with high agency costs and, therefore, expected to be associated with additional agency costs during the crisis period. The coefficient estimate on the interaction variable is negative when AT and EBITD are the dependent variables, suggesting an additional decline in asset turnover and earnings in public firms during the crisis period, but is statistically significant only in the former case. Similarly, the coefficient estimate

²⁹We repeat our tests including fiscal years 2007 and 2009. Results are qualitatively similar.

on the interaction variable when OPEXP is the dependent variable suggests that operating expenses are higher among public firms by an additional 0.1 standard deviations during the crisis period.

Models M4-M6 of Table 13 present results of tests examining how vertical agency costs change among owner-managed and non-owner managed private firms during the financial crisis. Here we treat non-owner-managed private firms as the group hypothesized to have higher agency costs compared to owner-managed private firms. Accordingly we define a dummy variable (NOM) that equals 1 if managerial ownership is less than 99% and 0 otherwise. Coefficient estimates on the interaction variable, our main variable of interest, in all three specifications suggest that non-owner-managed firms are associated with additional agency costs compared to owner-managed firms during the financial crisis.

Finally, Models M7-M9 of Table 13 present results of tests examining how horizontal agency costs change among firms with and without contestability in control during the financial crisis. We treat firms without contestability (No Contestability) - proxied by below median values of the second largest shareholder's stake - as firms hypothesized to have higher horizontal agency costs. Our results - specifically, the coefficient estimates on the interaction variable - again show that firms without contestability (hypothesized to have higher horizontal agency costs) respond more negatively to the financial crisis.

To summarize, the extant literature (Mitton, 2002; Lemmon and Lins, 2003) documents that, as expected, high agency cost firms do worse than low agency cost firms during an exogenous shock in the form of a financial crisis. In this context, the results presented in this section show that the direction of the incremental change during the financial crisis is consistent with the following: (a) public firms have higher vertical agency costs relative to private firms, (b) non-owner managed firms have higher vertical agency costs relative to owner managed firms, and (c) no-contestability firms have higher horizontal agency costs relative to firms with contestability in ownership structure. Hence, these results provide strong support to our bottom-line conclusions in the earlier sections.

7 Executive Compensation based Analyses of Agency Costs

We also use a non-accounting based measure as a proxy for agency costs in the spirit of Hartzell and Starks (2003), who use institutional ownership as a proxy for effective monitoring and executive compensation as a channel through which agency costs may be manifested.³⁰ They find that the level of compensation is negatively related to institutional ownership concentration while the pay-for-performance sensitivity of executive compensation is positively related to institutional ownership concentration. Hartzell and Starks (2003) conclude that institutions serve a monitoring role in mitigating the agency problem between shareholders and managers. In a similar spirit, we examine the relation between executive compensation and proxies for monitoring in our sample.

7.1 Tests Using Level of Compensation

An aspect of the compensation structure in which monitoring plays a role is the magnitude of compensation. If the various monitoring mechanisms examined so far (i.e., private firms over public firms; owner-managed firms over non-owner-managed firms etc.) are effective, then we would expect firms associated with effective monitoring to have managerial compensation no greater than what would be expected after controlling for economic determinants of the level of executive compensation. In contrast agency problems could manifest as abnormal positive deviations of actual compensation from what would otherwise be normal. Extant theory and empirical work suggests that firms with greater growth opportunities and complex operations pay higher wages to their executives (Core et al., 1999; Smith and Watts, 1992; Conyon et al., 2011). These results serve as our basis for identifying variables that are predicted to determine the normal level of compensation for a firm. We proxy for firm size and complexity with total assets. We proxy for investment opportunities using lagged sales growth rate as data required to

³⁰In a related study Core, Holthausen, and Larcker (1999) find that CEOs earn greater compensation when governance structures are less effective. See also Almazan, Hartzell, and Starks (2005) and Conyon, Core, and Guay (2011).

calculate market valuation measures are unavailable for our sample of private firms. We proxy for past performance with operating profit scaled by assets. We also control for leverage, industry and time effects. Finally we include managerial ownership as an explanatory variable. Fernandes, Ferreira, Matos, and Murphy (2013) document a negative relation between executive compensation and insider ownership and interpret the result as evidence that the two are substitutes. We follow the methods of Hartzell and Starks (2003) and estimate the following specification to examine the relation between monitoring and agency problems:

$$\begin{aligned}
\text{Compensation Measure}_{it} = & \alpha + \beta_{MI} * \text{Monitoring Indicator}_{it-1} + \gamma_1 * \Delta \text{Op. Profit}_{it-1} + \\
& \gamma_2 * \Delta \text{Op. Profit}_{it} + \gamma_{MO} * \text{Managerial Ownership}_{it-1} + \\
& \gamma_{Size} * \text{Assets}_{it} + \gamma_{Growth} * \text{Sales Growth}_{it-1} + \gamma_{Firm} * \text{Firm Controls}_{it} \\
& + \delta_{Industry} * \text{Industry Dummies} + \delta_t * \text{Year Dummies} + \varepsilon_{it}, \quad (7)
\end{aligned}$$

where "Monitoring Indicator" is the main variable of interest and varies depending on the sample used and the control variables act as determinants of what would otherwise be the normal level of compensation. For instance, in tests comparing owner-managed and non owner-managed firms, "Monitoring Indicator" is a dummy variable that equals 1 for owner-managed firms and 0 otherwise. Appendix B presents a detailed description of this variable. The coefficient estimate on "Monitoring Indicator" represents the excess compensation associated with effective monitoring after controlling for economic determinants of compensation. A non-positive coefficient estimate on "Monitoring Indicator" implies firms associated with effective monitoring do not overpay executives. We use two compensation measures. To examine the relation between monitoring and the vertical agency problem, we use total compensation paid to all managers as our compensation measure. To examine the relation between monitoring and the horizontal agency problem, we limit our sample to private firms with more than two shareholders and for which at least 75% of the ownership stake is accounted for. We use the compensation of the highest paid manager as the compensation measure as our objective is to

examine the role of monitoring on agency costs *amongst* owners. Since our sample is dominated by private companies, we do not have information on bonuses, stock options and other incentive payout structures.

Results are presented in Table 14. Model M1 presents results using the full sample of public and private firm-year observations. "Private" is a dummy variable that equals 1 for private firms and 0 otherwise. The coefficient estimate on "Private" is negative and significant suggesting private firms on average pay lower compensation to executives that is in line with what would be normal. Model M2 presents results with manager ownership as an additional explanatory variable. We find that executive compensation is negatively related to manager ownership stake but more importantly, the coefficient estimate on "Private" still remains negative and significant. Model M3 presents results using the sample of private firms. If owner-managed firms are associated with lower agency costs and better monitoring, we expect the coefficient estimate on "Owner-Managed" to be non-positive. We find that the coefficient estimate is negative and significant at the 10% level. The main variable of interest in model M4 is a dummy variable labeled "Few Shareholders" that equals 1 if the number of shareholders is less than 5 (proxying for simpler ownership structures) and 0 otherwise. The coefficient estimate is negative and significant suggesting that executive compensation is on average \$95,000 less in firms with simpler ownership structures compared to firms with more complex ownership structures. Models M5 and M6 present results of specifications designed to examine the role of monitoring on agency costs amongst owners. The dependent variable in these specifications is the compensation of the highest paid director. The coefficient estimates on "Low Concentration" are negative and statistically significant indicating that the compensation of the highest paid director is lower in firms with shared control. Similarly, the coefficient estimate on "Second Largest Stake" is negative and significant indicating a negative relation between the compensation of highest paid director and contestability.³¹

³¹We also use highest paid manager's compensation divided total executive compensation as the dependent variable in models M5 and M6. Results are qualitatively similar and are available on request

7.2 Tests Using Changes in Compensation:

In this section, we examine how sensitive executive compensation is to changes in performance (Jensen and Murphy, 1990). Our tests on the relation between effective monitoring and this sensitivity are based on the following equation:

$$\begin{aligned}\Delta Compensation Measure_{it} = & \alpha + \beta_{MI} * (Monitoring Indicator_{it-1} * \Delta Op. Profit_{it}) + \\ & \gamma_1 * \Delta Op. Profit_{it-1} + \gamma_{MO} * Managerial Ownership_{it-1} + \\ & \gamma_{Growth} * Sales Growth_{it-1} + \gamma_{Firm} * Firm Controls_{it} + \\ & \delta_{Industry} * Industry Dummies + \delta_t * Year Dummies + \varepsilon_{it}, (8)\end{aligned}$$

The coefficient estimate on the interaction variable measures the incremental effect of effective monitoring on the pay-for-performance sensitivity of executive compensation. A positive and significant estimate of the coefficient on the interaction variable indicates a greater alignment of executive compensation with past and current performance in firms with effective monitoring relative to firms without such monitoring. Table 15 presents the results. Model M1 presents results comparing public and private firms. The interaction variable is insignificant indicating no substantial difference in pay-for-performance sensitivity between public and private firms. In tests using various samples of private firms however, we find that the interaction variable is positive and significant in specifications testing the role of owner-managed firms (model M2), complexity of ownership structures (model M3), shared control (model M4) and contestability (model M5) as effective monitors that help reduce vertical and horizontal agency costs respectively.

8 Robustness Tests: Large Sample Analysis

We had hitherto restricted our sample firms to those that are not exempt from reporting and auditing requirements of the Companies House. As acknowledged earlier, a potential concern

with this criterion is the possible sample selection bias from imposing the size threshold implied by the auditing requirements of the Companies House. For instance, it could be argued that such a restriction might result in a sample that includes only large, successful, single owner-managed firms but includes poorly performing firms with other ownership structures. Given that a majority of our agency cost proxies are accounting variables, this selection bias maybe driving our results. Hence, to address any such concerns, we repeat our analysis by including private firms that are exempt from auditing requirements (i.e., a sample of 500,087 firm-year observations from 217,025 unique firms spanning fiscal years 2002-2010), and find that our conclusions and findings are not influenced by these size criteria. Note that the matched sample analysis presented so far should not be affected by this change as all our matching routines include size as a matching criterion. Hence, we repeat the regressions not using matched samples with the expanded dataset and find that our bottom-line findings are totally robust to the inclusion of small private firms. Results are not presented for brevity and are available on request.

9 Summary and Conclusions

The economic theory of agency differentiates between vertical agency problems and horizontal agency problems. The vertical agency problem is characterized by situations in which the managers of a firm have incentives to use or consume resources in a fashion that is at odds with the objectives of the firm's owners. The horizontal agency problem on the other hand arises when the owners themselves vie for resources, and importantly situations in which a controlling owner has incentives to exploit minority owners. This study presents comprehensive empirical tests of the central hypotheses regarding the relation between agency costs and the structure of ownership and control in an organization. We present tests of predictions regarding vertical agency costs as well as horizontal agency costs and find support for the basic tenets of both theses. Our results are based upon a unique data set containing information on the ownership and management of private as well as publicly traded companies.

We find that private firms experience lower agency costs when compared to public firms. More importantly, we document that a firm that is owned and managed by a single individual has lower agency costs than firms with other ownership and management structures, thus providing direct confirmation of the prediction first laid out by Jensen and Meckling (1976). We also find that among private firms, agency costs increase as firms move from simple ownership structures such as being owned by a single individual or a family, to complicated ownership structures such as being owned by multiple families and holding companies. This result provides direct confirmation of the free-rider hypothesis which states that as the number and type of shareholders increase, the incentive for each shareholder to incur all of the monitoring costs decreases because the benefits associated with monitoring are limited and are proportional to their ownership stake.

We present new empirical evidence on horizontal agency problems, a major concern in private firms and close corporations. We find that firms with shared control (i.e., those where the largest stakeholder owns less than 50%) have lower agency costs than firms where the largest shareholder has enough power to extract private benefits from minority shareholders. Furthermore, we find that the presence of a large shareholder outside the controlling group leads to lower agency costs. We also document an inverse relationship between the ownership stake of the second largest shareholder and agency costs, providing support for the view that minority expropriation will be lower in companies where control is more contestable. Tests accounting for endogeneity support our general conclusions as do additional robustness checks.

Appendix A: Identification of the sample

Data are obtained from updates of the FAME database (produced by *Bureau Van Dijk*) dated October 2005, October 2006, November 2007, October 2008, October 2009 and October 2010.

Notes for Table A.3: We clean the data and classify firms into various ownership structures. For example, according to table A.2, there are 7,806 firms owned by individuals in year 2005. Of these firms, we were able to identify 1,066 firms as being owned by one individual, 2,167 firms as being owned by one family (all individuals with the same last name are considered part of a family) and 3,261 firms as being owned by multiple families. We lose observations because of unclear owner names, missing ownership stakes etc.

Notes for Table A.4: We merge the firms in Table A.3 with those in Table A.1. Furthermore we exclude firms in the financial and utility industries and also exclude firms with turnover less than \$50,000 as many data points are missing for firms smaller than this size threshold.

Bureau Van Dijk Data Update:	2005	2006	2007	2008	2009	2010
Table A.1: Initial screening criteria for company selection						
Firms with minimum assets of \$5,000 and Min. Sales of \$1,000 and classified as Private limited, Public AIM, Public Quoted OFEX, Public, Not Quoted and Public, Quoted:	35,463	280,728	279,270	269,060	246,039	222,016
Firms without missing SIC codes:	34,413	278,991	275,976	263,520	240,466	217,256
Firms with account types classified as one of the following - Full accounts, Group, Medium Company, Partial Exemption, Small company, Total exemption Full, Total Exemption Small:	34,329	278,270	274,074	261,753	238,936	216,594
Fiscal year 2002 and above:	32,008	277,466	273,677	261,370	238,779	216,404
Table A.2: Identifying and classifying lagged ownership data						
Firms with lagged ownership data:	51,765	280,593	279,110	268,971	245,948	221,846
Standalone firms (i.e., firms without holding companies):	18,375	222,109	217,770	208,176	185,711	160,314
Firms without missing shareholder names: <i>of which</i>	15,100	195,858	195,383	188,813	169,080	150,084
Firms owned by individuals:	7,806	173,234	173,277	165,288	146,830	129,289
Firms owned by combination of individuals and holding companies:	7,294	22,624	22,106	23,525	22,250	20,795
Table A.3: Further classification of ownership data:						
Private firms owned by one individual	1,066	56,571	59,904	59,146	52,156	45,919
Private firms owned by one family	2,167	56,256	55,644	51,739	43,570	37,251
Private firms owned by multiple families	3,261	50,438	57,313	54,078	43,753	39,002
Private firms owned by individuals and companies	4,059	16,570	19,532	20,863	16,138	14,844
Public Firms	1,010	1,421	1,557	1,431	1,255	1,199
Table A.4: Merging ownership data and financial information						
Private firms owned by one individual:	533	32,789	27,893	28,926	25,483	24,869
Private firms owned by one family:	1023	27,599	32,883	33,545	25,797	25,014
Private firms owned by multiple families:	1,847	25,132	33,041	34,863	22,809	22,855
Private firms owned by individuals and companies:	2,678	13,685	14,706	17,965	9,656	8,340
Public Firms:	458	1,052	1,266	1,201	1,090	1,089
Table A.5: Identifying firms that are not subject to exemptions from Companies House auditing and reporting requirements (Sample used for our analysis)						
Private firms owned by one individual	523	1,859	1,891	1,963	1,657	1,755
Private firms owned by one family	979	2,410	3,223	3,366	2,263	2,838
Private firms owned by multiple families	1,799	5,038	7,600	8,910	4,625	5,322
Private firms owned by individuals and companies	2,452	9,351	9,770	12,401	6,178	5,205
Public Firms	458	1,052	1,266	1,201	1,090	1,089

Appendix B: Following is the list of the variable descriptions.

Assets	Total assets, measured in thousands
Available Stake	Total ownership stake that could be accounted for
Bank Debt	Bank loans outstanding / Total assets
Bank Debt Dummy (0/1)	Dummy variable that equals 1 if the firm has a loan from bank and zero otherwise
Contestable (0/1)	Dummy variable that equals 1 for above median values of second largest shareholders' stake and zero otherwise
Crisis (0/1)	Dummy variable that equals 1 for fiscal year 2008 and zero otherwise
Difference	Difference between the ownership stakes of the largest and second largest shareholders
Dividend Dummy (0/1)	Dummy variable that equals 1 if the firm pays dividend and zero otherwise
EBIT	Earnings before interest and tax / Total assets
EBITD	Earnings before interest, tax, depreciation and amortization / Total assets
Few Shareholders (0/1)	Dummy variable that equals 1 if the number of shareholders is less than five and zero otherwise
Firm Age	Firm age measured in years from date of incorporation to statement date
Firm Controls	Firm controls include firm age, bank loan, total liabilities, number of subsidiaries, a dummy variable <i>group</i> that equals to 1 if group accounts are filed, a dummy variable <i>dividend dummy</i> that equals if firms pays a dividend, and dummy variables <i>stable band</i> , <i>normal band</i> , <i>caution band</i> and <i>high risk band</i> that equal 1 if the QuiScore of the company falls between 60-80, 40-60, 20-40 and below 20 respectively. Secure band (QuiScore between 80 and 100) is the excluded group. A lower score indicates higher likelihood of company failure
Group Accounts (0/1)	Dummy variable that equals 1 if the company files Group accounts and zero otherwise
Largest Stake	Ownership stake of largest shareholder
Low Concentration (0/1)	Dummy variable that equals 1 if the largest shareholder owns less than 50% of the firm and zero otherwise
Managerial ownership	Total ownership stakes of directors
Monitoring Indicator	Variable proxying for effective monitoring. In tests comparing public and private firms, monitoring indicator is <i>Private</i> . In tests comparing agency costs between owner-managed and non-owner-managed firms, monitoring indicator is <i>OM</i> . In tests comparing agency costs between firms with simple and complex ownership structures, monitoring indicator is <i>Few Shareholders</i> . In tests comparing agency costs between firms with and without minority shareholder expropriation, monitoring indicator <i>Low Concentration</i> . In tests comparing agency costs between firms with and without contestability of control, monitoring indicator is the ownership stake of second largest shareholder
No Contestability (0/1)	Dummy variable that equals 1 for below median values of second largest shareholders' stake and zero otherwise
NOM or Non Owner-Managed (0/1)	Dummy variable that equals 1 if managers collectively own less than 99% of the firm and zero otherwise
OM or Owner-Managed (0/1)	Dummy variable equals 1 if managers collectively own at least 99% of the firm and zero otherwise
Op. Profit	Operating profit scaled by total assets
Private (0/1)	Dummy variable equals 1 for private firms and zero otherwise
Public (0/1)	Dummy variable equals 1 for public firms and zero otherwise
Sales Growth	One-year growth rate in sales
Second Largest Stake	Ownership stake of the second largest shareholder
Total Comp	Total compensation of all managers (in thousands)
Top Exec Comp	Compensation of the highest paid manager (in thousands)
Total Debt	Total debt scaled by total assets
Total Liabilities	Total liabilities scaled by total assets

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Table 1: Distribution of the sample

Panel A presents distribution of the full sample (42,459 unique firms and 109,434 firm-year observations) and sub-samples of public firms and private firms. Private firms are further classified into firms owned by one individual, firms owned by one family, firms owned by multiple families and firms owned by individuals and companies respectively. OM refers to owner-managed firms where managers collectively own more than 99% of the firm.

Panel B presents industry distribution of the full sample and sub-samples of public firms and private firms. Classification of firms into industry groups is done according to the UK Standard Industrial Classification of Economic Activities 2003 [SIC(2003)]. Please refer to www.ons.gov.uk for further details.

Panel A: Yearly distribution of the sample											
Fiscal Year	Full sample	Public firms	Private firms	Firms owned by one individual		Firms owned by one family		Firms owned by multiple families		Firms owned by individuals and companies	
				N	OM	N	OM	N	OM	N	OM
2004*	7,056	462	6,594	768	597	1,311	91	2,381	630	2,163	11
2005	18,112	1,090	17,022	1,754	1,407	2,930	216	6,045	1,787	6,293	74
2006	22,189	1,222	20,967	1,920	1,503	3,016	211	7,011	1,912	9,020	115
2007	22,477	1,195	21,282	1,913	1,437	2,819	169	6,632	1,764	9,918	93
2008	21,963	1,099	20,864	1,856	1,396	2,650	153	6,144	1,617	10,214	85
2009**	17,737	1,088	16,649	1,451	1,088	2,353	133	5,116	1,345	7,749	68
Firm-years (Firms)	109,534 (42,459)	6,156 (1,761)	103,378 (40,698)	9,648 (4,657)	7,427 (3,644)	15,079 (6,375)	973 (411)	33,294 (13,741)	9,053 (4,074)	45,357 (18,733)	446 (195)

** includes observations from 2002 and 2003. ** includes observations from 2010.*

Panel B: Industry distribution of the sample							
One Digit SIC Code	Business Description	Full Sample		Public Firms		Private Firms	
		N	% of Sample	N	% of Sample	N	% of Sample
0	Agriculture, Hunting and Forestry; Fishing	444	1.05%	14	0.80%	433	1.06%
1	Manufacture of Food Products	901	2.12%	73	4.15%	840	2.06%
2	Manufacture of Chemicals and Chemical Products	3,012	7.09%	243	13.80%	2,813	6.91%
3	Manufacture of Other Transport Equipment	1,720	4.05%	184	10.45%	1,562	3.84%
4	Construction	3,432	8.08%	86	4.88%	3,360	8.26%
5	Wholesale and Retail Trade	7,841	18.47%	246	13.97%	7,628	18.74%
6	Transport, Storage and Communication	2,273	5.35%	120	6.81%	2,181	5.36%
7	Real Estate, Renting and Business Activities	18,438	43.43%	636	36.12%	17,611	43.27%
8	Education, Health and Social Work	1,181	2.78%	30	1.70%	1,158	2.85%
9	Other Business Activities	3,217	7.58%	129	7.33%	3,112	7.65%
Total		42,459		1,761		40,698	

Table 2: Summary statistics of variables

Panel A presents distributional characteristics of the main variables of interest for the 103,378 private firm-year observations and 6,156 public firm-year observations in the sample. Panel B presents the mean and median values of main variables of interest for various sub-samples of private firm-year observations classified according to ownership structure. Please refer to Appendix B for a description of the variables.

Panel A												
Variable	Private Firms						Public Firms					
	N	Mean	Q25	Median	Q75	STD	N	Mean	Q25	Median	Q75	STD
<i>Financial Variables</i>												
Asset Turnover	103,378	2.059	0.614	1.573	2.717	2.163	6,156	1.121	0.472	0.915	1.464	1.053
Assets (000's)	103,378	32,498	1,152	4,141	11,847	764,028	6,156	906,991	10,226	44,973	226,291	5,719,597
Sales (000's)	103,378	24,239	873	4,880	15,303	193,653	6,156	708,016	7,051	35,769	224,425	4,204,324
Firm Age (years)	103,378	17.2	4.9	11.0	22.6	18.2	6,156	25.8	5.9	12.0	29.2	30.7
Total Debt	93,770	0.416	0.080	0.265	0.568	0.544	5,681	0.237	0.059	0.174	0.319	0.301
Total Liabilities	103,326	0.757	0.449	0.696	0.909	0.606	6,156	0.568	0.343	0.522	0.695	0.431
Bank Debt	91,194	0.063	0.000	0.002	0.069	0.126	5,567	0.063	0.002	0.023	0.073	0.109
Bank Loan (0/1)	103,378	0.425	0.000	0.000	1.000	0.494	6,156	0.652	0.000	1.000	1.000	0.476
OPEXP	103,378	0.463	0.143	0.287	0.619	0.557	6,156	0.711	0.214	0.418	0.861	0.891
EBITD	88,942	0.093	0.034	0.092	0.179	0.289	6,091	-0.014	-0.037	0.075	0.137	0.315
Dividend Dummy	103,378	0.311	0.000	0.000	1.000	0.463	6,156	0.504	0.000	1.000	1.000	0.500
Total Comp (000's)	23,090	476	115	265	539	875	2,715	1,410	478	864	1,726	1,786
Top Exec Comp (000's)	12,987	317	119	180	305	618	2,642	516	186	331	638	562
Qui Score	100,989	76	67	83	91	18	5,960	77	68	85	92	20
<i>Ownership Variables</i>												
Available Stake	103,374	0.991	1.000	1.000	1.000	0.073	6,150	0.720	0.511	0.816	1.000	0.301
Managerial Ownership	70,508	0.573	0.286	0.500	0.990	0.334	1,314	0.174	0.015	0.085	0.247	0.216
Owner Managed (0/1)	103,378	0.173	0.000	0.000	0.000	0.378	6,156	0.003	0.000	0.000	0.000	0.051
Largest Stake	103,374	0.654	0.429	0.570	1.000	0.288	6,150	0.206	0.105	0.160	0.254	0.160
Second Largest Stake	103,374	0.199	0.000	0.189	0.333	0.186	6,150	0.113	0.065	0.100	0.143	0.075
Panel B												
	Firms owned by one individual			Firms owned by one family			Firms owned by multiple families			Firms owned by individuals and companies		
	N	Mean	Median	N	Mean	Median	N	Mean	Median	N	Mean	Median
<i>Financial Variables</i>												
Asset Turnover	9,648	2.320	1.627	15,079	2.058	1.623	33,294	2.174	1.813	45,357	1.920	1.374
Assets (000's)	9,648	15,041	3,292	15,079	11,793	4,359	33,294	12,786	3,886	45,357	57,565	4,697
Sales (000's)	9,648	16,126	3,522	15,079	14,848	5,632	33,294	15,601	5,784	45,357	35,427	4,128
Firm Age (years)	9,648	13.5	9.3	15,079	21.8	18.4	33,294	20.2	14.3	45,357	14.3	7.1
Total Debt	8,581	0.399	0.273	14,051	0.299	0.220	30,182	0.300	0.194	40,956	0.546	0.363
Total Liabilities	9,641	0.765	0.731	15,079	0.617	0.612	33,281	0.672	0.660	45,325	0.864	0.760
Bank Debt	8,354	0.076	0.004	13,813	0.073	0.015	29,582	0.068	0.010	39,445	0.054	0.000
Bank Loan (0/1)	9,648	0.435	0.000	15,079	0.543	1.000	33,294	0.486	0.000	45,357	0.338	0.000
OPEXP	9,648	0.429	0.265	15,079	0.350	0.232	33,294	0.398	0.263	45,357	0.554	0.345
EBITD	8,240	0.120	0.096	13,876	0.129	0.099	30,317	0.120	0.098	36,509	0.052	0.082
Dividend Dummy	9,648	0.301	0.000	15,079	0.418	0.000	33,294	0.399	0.000	45,357	0.213	0.000
Total Comp (000's)	1,842	356	172	6,344	330	164	14,761	414	248	18,913	714	361
Top Exec Comp (000's)	896	347	176	2,680	299	174	7,647	275	163	12,580	392	213
Qui Score	9,532	75	81	14,684	79	88	32,787	79	88	43,986	75	80
<i>Ownership Variables</i>												
Available Stake	9,648	0.997	1.000	15,079	0.999	1.000	33,294	0.999	1.000	45,353	0.982	1.000
Managerial Ownership	7,459	0.998	1.000	13,749	0.440	0.500	30,845	0.646	0.667	18,455	0.378	0.333
Owner Managed (0/1)	9,648	0.770	1.000	15,079	0.065	0.000	33,294	0.272	0.000	45,357	0.010	0.000
No. of Shareholders	9,648	1.000	1.000	15,079	2.646	2.000	33,294	4.568	4.000	45,353	4.608	2.000
Largest Stake	9,648	0.997	1.000	15,079	0.614	0.500	33,294	0.510	0.500	45,353	0.699	0.755
Second Largest Stake	9,648	0.000	0.000	15,079	0.308	0.333	33,294	0.279	0.260	45,353	0.147	0.098

Table 3: Agency costs in public and private companies: Matched sample analysis

This table presents results of matched-sample analysis of public firms (treatment group) and private firms (control group). Matching criteria for sample presented in Panel A is as follows: We limit our sample to firms with non-missing values of asset turnover ratio, operating expenses to sales ratio (OPEXP) and earnings before interest, tax and depreciation scaled by total assets (EBITD). There are 6,091 public firm-year observations and 88,942 private firm-year observations that meet this criteria. Each observation in the treatment group is first matched with observations in the control group belonging to the same fiscal year, same two-digit SIC code and size within 30% of total assets. Among the potential matches, we identify an observation with a similar likelihood of company failure and closest in total debt.

Matching criteria for sample presented in Panel B are similar to the description except that our sample is limited a balanced panel and matching is only done in fiscal year 2005. A balanced panel is created by limiting sample period to 2005 to 2009 and including only those firms that are a.) not associated with changes in ownership structure during the period and b.) show up in the sample period every year. Using 489 public firms and 3,323 private firms in fiscal year 2005, we were able to find matches for 313 public firms. These matched firms are then followed from 2005-2009 resulting in 1,565 matched firm-year observations.

Panel C through Panel H present results of matched sample analysis by fiscal year.

One sample t-test (wilcoxon sign rank test) is employed to test the null hypothesis that the mean of (median of) paired differences is zero. Statistical significance at 10, 5 and 1 percent levels are respectively denoted by *, ** and ***. Please refer to Appendix B for a description of the variables.

		Treatment Group: Public firms		Control Group: Private firms		Paired Differences			
Variable	N	Mean	Median	Mean	Median	Mean	T-stat	Median	Z-stat
Panel A: Matches based on unbalanced panel									
Asset Turnover	5,163	1.148	0.932	1.354	1.040	-0.205***	-9.31	-0.071***	-8.45
OPEXP	5,163	0.747	0.432	0.514	0.319	0.233***	17.05	0.080***	17.32
EBITD	5,163	-0.033	0.063	0.086	0.089	-0.119***	-23.00	-0.043***	-23.08
Panel B: Matches based on balanced panel									
Asset Turnover	1,565	1.169	1.013	1.631	1.246	-0.462***	-9.52	-0.142***	-9.06
OPEXP	1,565	0.701	0.421	0.451	0.300	0.250***	11.16	0.100***	10.90
EBITD	1,565	0.013	0.078	0.112	0.102	-0.099***	-14.05	-0.042***	-13.34
Panel C: Matches based on sample of firms in fiscal year 2004									
Asset Turnover	335	1.018	0.852	1.206	0.903	-0.188***	-2.04	-0.024	-1.27
OPEXP	335	0.322	0.255	0.289	0.195	0.032	1.36	0.038***	3.16
EBITD	335	-0.102	0.009	0.018	0.042	-0.121***	-5.99	-0.040***	-5.52
Panel D: Matches based on sample of firms in fiscal year 2005									
Asset Turnover	893	0.946	0.760	1.238	0.890	-0.291***	-5.91	-0.115***	-5.31
OPEXP	893	0.548	0.344	0.383	0.257	0.165***	6.50	0.073***	7.07
EBITD	893	-0.128	0.008	0.001	0.054	-0.130***	-8.85	-0.057***	-10.03
Panel E: Matches based on sample of firms in fiscal year 2006									
Asset Turnover	1,067	1.027	0.806	1.209	0.948	-0.182***	-4.12	-0.073***	-4.05
OPEXP	1,067	0.671	0.402	0.513	0.318	0.157***	5.35	0.046***	5.70
EBITD	1,067	-0.093	0.035	0.054***	0.073	-0.148***	-12.20	-0.064***	-13.16
Panel F: Matches based on sample of firms in fiscal year 2007									
Asset Turnover	1,034	1.130	0.924	1.311	1.011	-0.181***	-3.86	-0.065***	-3.29
OPEXP	1,034	0.771	0.468	0.516	0.328	0.254***	8.16	0.082***	8.50
EBITD	1,034	-0.027	0.068	0.097	0.096	-0.124***	-11.33	-0.045***	-10.92
Panel G: Matches based on sample of firms in fiscal year 2008									
Asset Turnover	941	1.230	1.020	1.410	1.121	-0.178***	-3.75	-0.072***	-3.69
OPEXP	941	0.875	0.528	0.595	0.369	0.279***	8.10	0.105***	7.90
EBITD	941	0.022	0.093	0.139	0.119	-0.116***	-10.08	-0.039***	-9.58
Panel H: Matches based on sample of firms in fiscal year 2009									
Asset Turnover	869	1.450	1.200	1.688	1.340	-0.232***	-3.47	-0.051***	-2.64
OPEXP	869	1.039	0.599	0.648	0.412	0.391***	9.61	0.131***	8.85
EBITD	869	0.094	0.124	0.169	0.140	-0.074***	-6.99	-0.016***	-5.94

Table 4: Agency costs in public and private companies: Regression Analysis

This table presents regression results comparing agency costs in private and public companies. In each sample, all variables except dummy variables are standardized at the industry (two-digit SIC code) and fiscal year level. Panel A presents results of regressions using a matched sample of public and private firms. Models M1-M3 in Panel B present results using a matched sample created from a balanced panel. Models M4-M6 present results using the full sample. Please refer to the legend in Table 3 for a description of the matching criteria.

Public is a dummy variable that equals 1 if the firm is a listed on stock exchange. Please refer to Appendix B for definition of firm controls. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Model: Dependent Variable:	M1 AT_{it}	M2 $OPEXP_{it}$	M3 $EBITD_{it}$	M4 AT_{it}	M5 $OPEXP_{it}$	M6 $EBITD_{it}$
Panel A:						
$Public_{it-1}$	-0.107* (-2.002)	0.274*** (8.144)	-0.362*** (-20.967)	-0.160** (-2.064)	0.257*** (4.599)	-0.257*** (-4.989)
Managerial Ownership $_{it-1}$				0.038* (1.711)	-0.056*** (-2.955)	0.048*** (2.953)
$Public_{it-1}$ x Managerial Ownership $_{it-1}$				-0.025 (-0.464)	0.002 (0.032)	0.015 (0.253)
Assets $_{it}$	-0.174*** (-8.003)	-0.022 (-1.257)	0.041** (2.057)	-0.149*** (-4.276)	-0.084*** (-2.782)	0.050** (2.216)
Constant	0.112* (1.860)	-0.176*** (-3.654)	0.270*** (6.450)	0.414*** (3.754)	0.067 (0.669)	0.084* (1.695)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,359	9,359	9,359	3,876	3,876	3,876
Adjusted R-squared	0.12	0.09	0.18	0.16	0.10	0.16
Panel B:						
$Public_{it-1}$	-0.291* (-1.874)	0.185 (1.479)	-0.345*** (-2.680)	-0.270*** (-5.121)	0.398*** (3.872)	-0.250*** (-6.099)
Managerial Ownership $_{it-1}$	-0.025 (-0.349)	-0.141*** (-2.915)	0.006 (0.095)	0.051*** (9.954)	-0.024*** (-3.052)	0.046*** (7.245)
$Public_{it-1}$ x Managerial Ownership $_{it-1}$	0.102 (0.546)	0.087 (0.705)	0.059 (0.262)	-0.068* (-1.684)	0.009 (0.122)	0.000 (0.011)
Assets $_{it}$	-0.118 (-1.626)	-0.158** (-2.532)	0.098* (1.840)	-0.054*** (-2.853)	-0.064** (-2.172)	0.006 (0.740)
Constant	0.445*** (3.003)	0.008 (0.048)	0.572*** (4.215)	-0.056 (-0.751)	-0.105 (-0.967)	0.156 (1.269)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year Controls $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,149	1,149	1,149	64,346	64,346	59,621
Adjusted R-squared	0.19	0.14	0.24	0.13	0.04	0.11

Table 5: Agency costs in owner-managed and non owner-managed firms: Matched sample analysis

This table presents results of matched sample analysis of owner-managed (treatment group) and non owner-managed (control group) firms. Owner-managed firms (OM firms) are firms where managers collectively own more than 99% of the firm. Non owner-managed firms (Non OM firms) are firms where managers collectively own less than 99% of the firm. Matching criteria for sample presented in Panel A is as follows: We limit our sample to private firm-year observations with non-missing values of asset turnover ratio, operating expenses to sales ratio (OPEXP) and earnings before interest, tax and depreciation scaled by total assets (EBITD). We then randomly select 5,000 observations from 72,919 firm-year observations in the treatment group. Each observation in the treatment group is first matched with observations in the control group (there are 16,023 observations in this group) belonging to the same fiscal year, same two-digit SIC code and size within 30% of total assets. Among the potential matches, we identify an observation with a similar likelihood of company failure and closest in total debt.

Matching criteria for sample presented in Panel B are similar to the description except that our sample is limited to a balanced panel and matching is only done in fiscal year 2005. A balanced panel is created by limiting sample period to 2005 to 2009 and including only those firms that are a.) not associated with changes in ownership structure during the period and b.) show up in the sample period every year. Using 2,659 non owner-managed firms and 664 owner-managed firms in fiscal year 2005, we were able to find matches for 2,130 non owner-managed firms. These matched firms are then followed from 2005-2009 resulting in 10,650 matched firm-year observations.

Panel C presents results of matched sample analysis using a sample of firms owned by one individual. Matching criteria is similar to the description presented above for Panel A. The treatment group (owner-managed firms) in this sub-sample are closest to the description of the zero-agency cost firms described by Jensen and Meckling (1986).

One sample t-test (wilcoxon sign rank test) is employed to test the null hypothesis that the mean of (median of) paired differences is zero. Statistical significance at 10, 5 and 1 percent levels are respectively denoted by *, ** and ***. Please refer to Appendix B for a description of the variables.

		Treatment Group: Non OM firms		Control Group: OM firms		Paired Differences			
Variable	N	Mean	Median	Mean	Median	Mean	T-stat	Median	Z-stat
Panel A: Matches based on unbalanced panel									
Asset Turnover	4,601	2.071	1.657	2.200	1.810	-0.133***	-3.89	-0.061***	-5.15
OPEXP	4,601	0.477	0.304	0.388	0.261	0.089***	9.53	0.029***	8.74
EBITD	4,601	0.085	0.088	0.137	0.109	-0.052***	-10.29	-0.018***	-9.63
Panel B: Matches based on balanced panel									
Asset Turnover	10,650	2.064	1.799	2.212	1.965	-0.148***	-7.14	-0.070***	-7.85
OPEXP	10,650	0.385	0.256	0.351	0.225	0.034***	7.15	0.016***	9.35
EBITD	10,650	0.120	0.097	0.131	0.105	-0.010***	-5.53	-0.006***	-6.37
Panel C: Matches based on firms owned by one individual									
Asset Turnover	1,474	2.310	1.686	2.403	1.918	-0.092	-1.23	-0.056**	-2.00
OPEXP	1,474	0.460	0.307	0.400	0.255	0.059***	3.74	0.031***	4.19
EBITD	1,474	0.096	0.086	0.135	0.105	-0.038***	-4.10	-0.009***	-2.92

Table 6: Agency costs in owner-managed and non owner-managed firms: Regression analysis

This table presents regression results comparing agency costs among owner-managed and non owner-managed firms. Owner-managed firms are firms where managers collectively own more than 99% of the firm. Non owner-managed firms are firms where managers collectively own less than 99% of the firm. Panel A presents results of regressions using a matched sample of owner-managed and non owner-managed firms. Models M1-M3 in Panel B present results using a matched sample created from a balanced panel. Models M4-M6 in Panel B present results using the full sample of private firms. Please refer to the legend of Table 5 for a description of the matching criteria. In each sample, all variables except dummy variables are standardized at the industry (two-digit SIC code) and fiscal year level.

Owner-Managed is a dummy variable that equals 1 if managers collectively own more than 99% of the firm and 0 otherwise. Please refer to Appendix B for a description of firm controls and other variables. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Model:	M1	M2	M3	M4	M5	M6
Dependent Variable:	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$
Panel A:						
Owner-Managed $_{it-1}$	0.071*** (3.382)	-0.130*** (-5.016)	0.102*** (5.094)			
Managerial Ownership $_{it-1}$				0.032** (2.660)	-0.041*** (-2.986)	0.036*** (3.761)
Assets $_{it}$	-0.125*** (-5.437)	-0.094*** (-6.203)	-0.011 (-0.967)	-0.133*** (-6.242)	-0.114*** (-6.520)	-0.005 (-0.329)
Constant	-0.040* (-1.756)	-0.084** (-2.132)	0.076** (2.360)	0.126*** (4.860)	-0.066 (-1.429)	0.174*** (7.160)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,350	8,350	8,350	6,993	6,993	6,993
Adjusted R-squared	0.10	0.07	0.11	0.11	0.05	0.08
Panel B:						
Owner-Managed $_{it-1}$	0.111*** (2.807)	-0.108*** (-3.259)	0.056 (1.249)	0.096*** (6.364)	-0.063*** (-3.290)	0.103*** (11.814)
Assets $_{it}$	-0.083** (-2.406)	-0.124*** (-8.289)	0.016 (0.793)	-0.057*** (-5.568)	-0.018 (-1.410)	-0.001 (-0.156)
Constant	-0.074* (-1.742)	-0.053** (-2.489)	0.120*** (3.401)	-0.109 (-1.161)	-0.149 (-1.187)	0.221* (1.768)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,700	12,700	12,700	91,150	91,150	80,341
Adjusted R-squared	0.10	0.04	0.07	0.11	0.06	0.15

Table 7: Agency costs and complexity of ownership structure: Matched sample analysis

This table presents results of matched sample analysis of firms with complex ownership structure (treatment group) proxied by higher number of shareholders and firms with simple ownership structure (control group) proxied by fewer shareholders. Matching criteria for sample presented in Panel A is as follows: We limit our sample to private firms with at least 90% of ownership stake is accounted for and non-missing values of asset turnover ratio, operating expenses to sales ratio (OPEXP) and earnings before interest, tax and depreciation scaled by total assets (EBITD). We then randomly select 5,000 observations from 17,900 firm-year observations in the treatment group. Each observation in the treatment group is first matched with observations in the control group (there are 42,046 observations in this group) belonging to the same fiscal year, same two-digit SIC code and size within 30% of total assets. Among the potential matches, we identify an observation with a similar likelihood of company failure and closest in total debt. Matching criteria for sample presented in Panel B are similar to the description except that our sample is limited to a balanced panel and matching is only done in fiscal year 2005. A balanced panel is created by limiting sample period to 2005 to 2009 and including only those firms that are a.) not associated with changes in ownership structure during the period and b.) show up in the sample period every year. Using 705 private firms with more than five shareholders and 1,342 private firms with less than or equal to two shareholders in fiscal year 2005, we were able to find 631 matches. These matched firms are then followed from 2005-2009 resulting in 3,155 matched firm-year observations. Panels C and D present results of matched samples accounting for the effects of managerial ownership. More specifically, the matching criteria is similar to the criteria used for the sample in Panel A but the samples are limited to firms with above median and below median values of managerial ownership respectively. One sample t-test (wilcoxon sign rank test) is employed to test the null hypothesis that the mean of (median of) paired differences is zero. Statistical significance at 10, 5 and 1 percent levels are respectively denoted by *, ** and ***. Please refer to Appendix B for a description of the variables.

		Treatment Sample: Number of shareholders > 5		Control Sample: Number of shareholders <= 2		Paired Differences			
Variable	N	Mean	Median	Mean	Median	Mean	t-value	Median	Z-stat
Panel A: Matches based on unbalanced panel									
Asset Turnover	4,854	1.831	1.495	1.860	1.500	-0.029	-0.98	0.000	-0.75
OPEXP	4,854	0.484	0.311	0.430	0.270	0.053***	5.42	0.027***	7.30
EBITD	4,854	0.074	0.085	0.104	0.092	-0.029***	-7.06	-0.009***	-7.08
Panel B: Matches based on balanced panel									
Asset Turnover	3,155	1.875	1.656	2.034	1.829	-0.158***	-4.79	-0.098***	-5.41
OPEXP	3,155	0.399	0.270	0.358	0.248	0.040***	4.55	0.015***	5.19
EBITD	3,155	0.115	0.099	0.123	0.102	-0.008***	-2.37	-0.006***	-3.03
Panel C: Matches based on unbalanced panel and above median managerial ownership									
Asset Turnover	4,647	2.085	1.804	2.180	1.872	-0.095***	-3.02	-0.002*	-1.76
OPEXP	4,647	0.464	0.315	0.368	0.245	0.096***	11.15	0.049***	14.77
EBITD	4,647	0.099	0.101	0.126	0.106	-0.026***	-6.31	-0.004***	-4.06
Panel D: Matches based on unbalanced panel and below median managerial ownership									
Asset Turnover	4,693	1.630	1.352	1.733	1.404	-0.102***	-3.66	-0.021***	-3.02
OPEXP	4,693	0.522	0.323	0.467	0.294	0.055***	5.11	0.026***	5.61
EBITD	4,693	0.061	0.080	0.093	0.088	-0.031***	-6.81	-0.011***	-7.44

Table 8: Agency costs and complexity of ownership structures: Regression analysis

This table presents results of regression analysis of firms with complex ownership structures proxied by higher number of shareholders and firms with simple ownership structures proxied by fewer shareholders. Models M1-M3 in Panel A present results of regressions using a matched sample of firms with and without complex ownership structures. Models M4-M6 in Panel A present results using full sample of private firms. Models M1-M3 (M4-M6) in Panel B present results of regressions using matched samples accounting for the effects of managerial ownership. More specifically, the matching criteria is similar to the criteria used for the sample in Panel A but the samples are limited to firms with above median and below median values of managerial ownership respectively. Please refer to the legend for Table 7 for a description of the matching criteria. In each sample, all variables except dummy variables are standardized at the industry (two-digit SIC code) and fiscal year level.

Please refer to Appendix B for a description of firm controls. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Model:	M1	M2	M3	M4	M5	M6
Dependent Variable:	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$
Panel A:						
Number of Shareholders $_{it-1}$	-0.006 (-0.361)	0.082*** (4.859)	-0.069*** (-5.263)	-0.031*** (-3.496)	0.046*** (4.915)	-0.045*** (-7.352)
Managerial Ownership $_{it-1}$	0.038** (2.616)	0.016 (1.214)	0.007 (0.389)	0.036*** (7.122)	-0.007 (-0.913)	0.028*** (4.458)
Assets $_{it}$	-0.145*** (-6.153)	-0.098*** (-3.279)	0.002 (0.108)	-0.056*** (-3.622)	-0.071*** (-4.562)	0.008 (1.078)
Constant	-0.058 (-1.234)	-0.137*** (-3.529)	0.059* (1.752)	-0.053 (-0.711)	-0.124 (-1.112)	0.163 (1.294)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,639	6,639	6,639	63,175	63,175	58,465
Adjusted R-squared	0.13	0.05	0.08	0.12	0.04	0.10
Panel B:						
Number of Shareholders $_{it-1}$	-0.029 (-1.303)	0.103*** (6.995)	-0.030 (-1.659)	-0.076*** (-4.564)	0.097*** (4.424)	-0.135*** (-5.576)
Managerial Ownership $_{it-1}$	0.011 (0.486)	-0.008 (-0.434)	0.017 (1.067)	0.024 (1.208)	-0.015 (-0.836)	0.009 (0.576)
Assets $_{it}$	-0.091*** (-2.871)	-0.106*** (-6.784)	0.011 (0.802)	-0.135*** (-4.697)	-0.091*** (-2.961)	0.023 (0.928)
Constant	-0.071* (-2.008)	-0.260*** (-6.573)	0.165*** (5.165)	0.057 (1.177)	-0.241*** (-3.944)	0.215*** (4.022)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,401	8,401	8,401	4,450	4,450	4,450
Adjusted R-squared	0.10	0.06	0.12	0.13	0.08	0.09

Table 9: Minority expropriation and shared control: Matched sample analysis

This table presents results of matched sample analysis of firms without shared control proxied by largest shareholder owning more than 50% (treatment group) of the firm and firms with shared control proxied by largest shareholder owning less than 50% of the firm (control group). Matching criteria for sample presented in Panel A is as follows: We limit our sample to private firms with at least 90% of ownership stake is accounted for and non-missing values of asset turnover ratio, operating expenses to sales ratio (OPEXP) and earnings before interest, tax and depreciation scaled by total assets (EBITD). We then randomly select 5,000 observations from 15,471 firm-year observations in the treatment group. Each observation in the treatment group is first matched with observations in the control group (there are 27,789 observations in this group) belonging to the same fiscal year, same two-digit SIC code and size within 30% of total assets. Among the potential matches, we identify an observation with a similar likelihood of company failure and closest in total debt.

Matching criteria for sample presented in Panel B are similar to the description except that our sample is limited to a balanced panel and matching is only done in fiscal year 2005. A balanced panel is created by limiting sample period to 2005 to 2009 and including only those firms that are a.) not associated with changes in ownership structure during the period and b.) show up in the sample period every year. Using 1,166 private firms where largest shareholder owns more than 50% of the firm and 691 private firms where the largest shareholder owns less than 50% of the firm in fiscal year 2005, we were able to find 630 matches. These matched firms are then followed from 2005-2009 resulting in 3,150 matched firm-year observations.

One sample t-test (wilcoxon sign rank test) is employed to test the null hypothesis that the mean of (median of) paired differences is zero. Statistical significance at 10, 5 and 1 percent levels are respectively denoted by *, ** and ***. Please refer to Appendix B for a description of the variables.

		Treatment Group: Largest Shareholder Stake >50%		Control Group: Largest Shareholder Stake <=50%		Paired Differences			
Variable	N	Mean	Median	Mean	Median	Mean	T-stat	Median	Z-stat
Panel A: Matches based on unbalanced panel									
Asset Turnover	4,820	2.007	1.700	2.059	1.737	-0.052*	-1.74	-0.020*	-1.97
OPEXP	4,820	0.431	0.281	0.423	0.278	0.008	0.95	-0.002	-0.09
EBITD	4,820	0.082	0.086	0.102	0.093	-0.019***	-4.35	-0.007***	-4.64
Panel B: Matches based on balanced panel									
Asset Turnover	3,150	2.109	1.868	2.040	1.757	0.068**	2.18	-0.001	1.11
OPEXP	3,150	0.372	0.254	0.355	0.241	0.016**	2.14	0.005	0.36
EBITD	3,150	0.117	0.099	0.122	0.100	-0.005	-1.55	-0.004**	-2.22

Table 10: Minority expropriation and shared control: Regression analysis

This table presents results of matched sample analysis of firms without shared control proxied by largest shareholder owning more than 50% of the firm and firms with shared control proxied by largest shareholder owning less than 50% of the firm. Models M1-M3 present results of regressions using a matched sample of firms with and without complex ownership structures. Models M4-M6 present results using full sample of private firms. In each sample, all variables except dummy variables are standardized at the industry (two-digit SIC code) and fiscal year level. Please refer to the legend in Table 9 for a description of the matching criteria.

Low Concentration is a dummy variable that equals 1 if the largest shareholder owns less than 50% of the firm and 0 otherwise. Please refer to Appendix B for a description of firm controls. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Variable:	M1	M2	M3	M4	M5	M6
Dependent Variable:	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$
Low Concentration $_{it-1}$	0.048* (1.695)	-0.025 (-1.155)	0.054** (2.209)	0.042*** (3.400)	-0.026 (-1.341)	0.044*** (3.352)
Managerial Ownership $_{it-1}$	0.075*** (6.415)	-0.036*** (-2.719)	0.043** (2.561)	0.077*** (12.647)	-0.034*** (-4.261)	0.055*** (6.521)
Assets $_{it}$	-0.117*** (-4.778)	-0.121*** (-4.335)	0.025* (2.009)	-0.075*** (-4.304)	-0.084*** (-4.919)	0.014 (1.284)
Constant	-0.036 (-0.912)	-0.119*** (-2.715)	0.152*** (2.766)	-0.114 (-1.649)	0.000 (0.008)	0.281*** (4.980)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,845	7,845	7,845	39,192	39,192	36,790
Adjusted R-squared	0.12	0.05	0.10	0.12	0.04	0.10

Table 11: Minority expropriation and contestability: Matched sample analysis

This table presents results of matched sample analysis of firms without contestability proxied by relatively low ownership stake of second largest shareholder (treatment group) and firms with contestability proxied by high ownership stake of second largest shareholder (control group). Matching criteria for sample presented in Panel A is as follows: We limit our sample to private firms with more than two shareholders, at least 90% of ownership stake is accounted for and non-missing values of asset turnover ratio, operating expenses to sales ratio (OPEXP) and earnings before interest, tax and depreciation scaled by total assets (EBITD). We then randomly select 5,000 observations from 32,605 firm-year observations in the treatment group. Each observation in the treatment group is first matched with observations in the control group (there are 31,591 observations in this group) belonging to the same fiscal year, same two-digit SIC code and size within 30% of total assets. Among the potential matches, we identify an observation with a similar likelihood of company failure and closest in total debt.

Matching criteria for sample presented in Panel B are similar to the description except that our sample is limited to a balanced panel and matching is only done in fiscal year 2005. A balanced panel is created by limiting sample period to 2005 to 2009 and including only those firms that are a.) not associated with changes in ownership structure during the period and b.) show up in the sample period every year. Using 1,271 private firms with below median values of second largest shareholder's ownership stake and 1,362 private firms with above median values of second largest shareholder's ownership stake in fiscal year 2005, we were able to find 1,244 matches. These matched firms are then followed from 2005-2009 resulting in 6,220 matched firm-year observations.

One sample t-test (wilcoxon sign rank test) is employed to test the null hypothesis that the mean of (median of) paired differences is zero. Statistical significance at 10, 5 and 1 percent levels are respectively denoted by *, ** and ***. Please refer to Appendix B for a description of the variables.

		Treatment Group: Below median second largest shareholder stake		Control Group: Above median second largest shareholder stake		Paired Differences			
Variable	N	Mean	Median	Mean	Median	Mean	T-stat	Median	Z-stat
Panel A: Matches based on unbalanced panel									
Asset Turnover	4,818	1.952	1.622	2.113	1.773	-0.160***	-5.20	-0.064***	-5.96
OPEXP	4,818	0.458	0.296	0.401	0.262	0.057***	6.27	0.022***	6.72
EBITD	4,818	0.087	0.088	0.121	0.098	-0.033***	-7.61	-0.008***	-6.18
Panel B: Matches based on balanced panel									
Asset Turnover	6,220	2.003	1.747	2.031	1.799	-0.027	-1.20	-0.018***	-2.94
OPEXP	6,220	0.387	0.263	0.349	0.242	0.037***	6.11	0.014***	6.34
EBITD	6,220	0.114	0.095	0.125	0.104	-0.010***	-4.23	-0.007***	-5.69

Table 12: Minority expropriation and contestability: Regression analysis

This table presents results of matched sample analysis of firms without contestability proxied by relatively low ownership stake of second largest shareholder and firms with contestability proxied by high ownership stake of second largest shareholder. Models M1-M3 present results of regressions using a matched sample of firms with and without contestability of ownership. Models M4-M9 present results using full sample of private firms with more than 2 shareholders. In each sample, all variables except dummy variables are standardized at the industry (two-digit SIC code) and fiscal year level. Please refer to the legend of Table 11 for a description of the matching criteria.

Contestable is a dummy variable that equals 1 for above median ownership stakes of the second largest shareholder and 0 otherwise. *Difference* is the difference in ownership stakes between the largest shareholder and the second largest shareholder. Please refer to Appendix B for a description of firm controls and other control variables. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, **, * and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Model:	M1	M2	M3	M4	M5	M6	M7	M8	M9
Dependent Variable:	AT _{it}	OPEXP _{it}	EBITD _{it}	AT _{it}	OPEXP _{it}	EBITD _{it}	AT _{it}	OPEXP _{it}	EBITD _{it}
Contestable _{it-1}	0.082*** (3.090)	-0.119*** (-5.344)	0.072*** (4.190)						
Second Largest Stake _{it-1}				0.032*** (4.911)	-0.037*** (-5.422)	0.030*** (3.808)			
Difference _{it-1}							-0.028*** (-3.816)	0.023** (2.477)	-0.024*** (-3.063)
Managerial Ownership _{it-1}	0.054*** (3.267)	-0.026* (-1.816)	0.059*** (3.608)	0.072*** (11.413)	-0.030*** (-3.594)	0.051*** (5.891)	0.077*** (12.806)	-0.035*** (-4.214)	0.055*** (6.496)
Assets _{it}	-0.140*** (-8.595)	-0.098*** (-5.936)	0.002 (0.125)	-0.075*** (-4.278)	-0.084*** (-5.014)	0.015 (1.303)	-0.075*** (-4.294)	-0.084*** (-4.942)	0.014 (1.295)
Constant	-0.610*** (-16.610)	0.396*** (8.621)	-0.845*** (-16.098)	-0.089 (-1.515)	-0.014 (-0.280)	0.307*** (7.453)	-0.087 (-1.481)	-0.016 (-0.320)	0.309*** (7.366)
Firm Controls _{it}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects _t	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,424	7,424	7,424	39,192	39,192	36,790	39,192	39,192	36,790
Adjusted R-squared	0.13	0.05	0.08	0.12	0.04	0.10	0.12	0.04	0.10

Table 13: Tests using financial crisis as an exogenous shock

This table presents results of a difference-in-difference analysis on how firms hypothesized to have high and low agency costs respond to the exogenous shock represented by the 2008 financial crisis. *Crisis* is a dummy variable that equals 1 for fiscal year 2008 and 0 otherwise. We exclude fiscal years 2007 and 2009 in all tests. Results presented in models M1-M3 compare how agency costs change for private and public firms. Here *Public* is a dummy variable that equals 1 for a public firm and 0 otherwise. Results presented in models M4-M6 compare how agency costs change for owner-managed and non owner-managed (NOM) firms. Here *NOM* is a dummy variable that equals 1 for a non-owner-managed firms and 0 otherwise. Results presented in model M7-M9 compare how agency costs change for firms with contestability and without contestability. Here *No Contestability* is a dummy variable that equals 1 for below median values of second largest shareholder's stake and 0 otherwise. Please refer to Appendix B for a definition of firm controls and other explanatory variables. All regressions control for industry fixed effects and standard errors are clustered at the industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Sample:	Public vs Private				OM vs NOM			Contestability vs No Contestability			
Model:	M1	M2	M3	M4	M5	M6	M7	M8	M9		
Dependent Variable:	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$	AT_{it}	$OPEXP_{it}$	$EBITD_{it}$		
$Public_{it-1}$	-0.212*** (-6.619)	0.290*** (5.084)	-0.365*** (-9.759)								
$Crisis_t$	0.029*** (3.268)	0.020*** (3.679)	-0.061*** (-14.921)	0.050*** (3.372)	-0.018 (-1.067)	-0.011*** (-2.892)	0.037*** (4.769)	0.012 (1.397)	-0.058*** (-7.474)		
$Public_{it-1} \times Crisis_t$	-0.040*** (-2.968)	0.095** (2.660)	-0.043 (-0.787)								
NOM_{it-1}				-0.094*** (-5.046)	0.050** (2.172)	-0.036*** (-3.133)					
$NOM_{it-1} \times Crisis_t$				-0.019 (-1.159)	0.044** (2.262)	-0.126*** (-13.869)					
No Contestability $_{it-1}$							-0.087*** (-4.326)	0.082*** (3.830)	-0.104*** (-3.954)		
No Contestability $_{it-1} \times Crisis_t$							-0.055** (-2.344)	0.119** (2.399)	-0.070** (-2.400)		
Assets $_it$	-0.038*** (-5.907)	-0.021** (-2.217)	0.001 (0.139)	-0.057*** (-6.300)	-0.025* (-1.901)	0.000 (0.080)	-0.072*** (-6.887)	-0.041*** (-2.789)	0.010 (1.442)		
Constant	-0.056*** (-4.668)	-0.030*** (-2.856)	-0.010 (-1.270)	0.011 (0.407)	-0.071*** (-3.741)	0.082*** (5.932)	-0.014 (-0.834)	-0.028** (-2.355)	0.018 (1.617)		
Observations	61,478	61,478	54,496	57,902	57,902	51,031	29,137	29,137	26,817		
Adjusted R-squared	0.11	0.05	0.15	0.11	0.05	0.15	0.10	0.04	0.10		

Table 14: Regression analysis of the level of executive compensation

This table presents regression results of the level of executive compensation. Models M1-M4 examine the relationship between level of executive compensation and vertical agency problems. Models M5 and M6 examine the relationship between level of executive compensation and horizontal agency problems. Results presented in models M1 and M2 are based on the full sample of public and private firm-year observations. Results presented in model M3 are based on the full sample of private firm-year observations. Results presented in model M4 are based on a sample of private firm-year observations with more than two shareholders. Results presented in models M5 and M6 are based on a sample of private firm-year observations with at least two shareholders and managers owning at least 50% of the firm. *Private* is a dummy variable that equals 1 for a private firm and 0 otherwise. *Owner-Managed* is a dummy variable that equals 1 if managers collectively own more than 99% of the firm and 0 otherwise. *Few shareholders* is a dummy variable if the firm has less than five shareholders and 0 otherwise. *Low Concentration* is a dummy variable that equals 1 if the largest shareholder's ownership stake is less than 50% and 0 otherwise. Please refer to Appendix B for a definition of firm controls and other explanatory variables. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Model:	M1	M2	M3	M4	M5	M6
Dependent Variable:	Total Executive Compensation _{it}				Highest Paid Exec. Compensation _{it}	
$\Delta \text{Op. Profit}_{it-1}$	-0.000** (-2.169)	0.001 (0.588)	0.004 (0.966)	0.006 (0.981)	0.004 (1.239)	0.004*** (11.086)
$\Delta \text{Op. Profit}_{it}$	-0.000 (-0.886)	0.000 (0.039)	0.001 (0.753)	0.003 (1.080)	0.002 (1.126)	0.002*** (7.279)
Private_{it-1}	-304.27*** (-6.241)	-121.28** (-2.617)				
$\text{Managerial Ownership}_{it-1}$		-46.13*** (-2.742)				
$\text{Owner-Managed}_{it-1}$			-46.72* (-1.667)			
$\text{Few Shareholders}_{it-1}$				-95.22*** (-4.112)		
$\text{Low Concentration}_{it-1}$					-59.07** (-2.341)	
$\text{Second Largest Stake}_{it-1}$						-151.25* (-1.952)
$\text{Sales Growth}_{it-1}$	0.0861*** (6.281)	1.140 (1.166)	0.073*** (4.618)	3.076 (0.910)	7.203 (0.915)	7.254** (2.496)
Assets_{it}	0.000*** (2.145)	0.001*** (4.679)	0.001*** (2.902)	0.001*** (2.736)	0.000*** (2.275)	0.000*** (19.843)
Constant	472.76*** (3.322)	198.47*** (3.350)	238.81* (1.715)	172.39 (0.654)	209.53 (0.782)	201.96 (0.666)
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects _t	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,069	18,334	21,589	13,294	3,781	3,780
Adjusted R-squared	0.28	0.13	0.13	0.18	0.17	0.17

Table 15: Regression analysis of the change in executive compensation

This table presents regression results of changes in executive compensation. The dependent variable in models M1 through M3 is the change total executive compensation. The dependent variable in models M4 and M5 is the change in compensation of highest paid executive. Results presented in M1 are based on the full sample of public and private firm-year observations. Results presented in M2 are based on the full sample of private firm-year observations. Results presented in M3 are based on a sample of private firm-year observations with more than two shareholders. Results presented in M4 and M5 are based on a sample of private firm-year observations with at least two shareholders and managers owning at least 50% of the firm.

Private is a dummy variable that equals 1 for a private firm and 0 otherwise. *Owner-Managed* is a dummy variable that equals 1 if managers collectively own more than 99% of the firm and 0 otherwise. *Few shareholders* is a dummy variable if the firm has less than five shareholders and 0 otherwise. *Low Concentration* is a dummy variable that equals 1 if the largest shareholder's ownership stake is less than 50% and 0 otherwise. Please refer to Appendix B for definition of firm controls and other variables. All regressions control for year fixed effects and industry fixed effects. Standard errors are clustered at the year-industry level. t-values are reported in parentheses. ***, ** and * stand for statistical significance based on a two-sided t test at the 1%, 5% and 10% level, respectively.

Model: Dependent Variable:	M1	M2 Δ Total Executive Compensation $_{it}$	M3	M4 Δ Highest Paid Exec. Compensation $_{it}$	M5
Δ Op. Profit $_{it-1}$	-0.000*** (-3.875)	0.003 (0.902)	0.004*** (12.972)	0.005*** (13.923)	0.004*** (15.050)
Private $_{it-1}$ x Δ Op. Profit $_{it}$	-0.000 (-0.115)				
Private $_{it-1}$	-30.416 (-1.555)				
Owner-Managed $_{it-1}$ x Δ Op. Profit $_{it}$		0.027** (2.044)			
Owner-Managed $_{it-1}$		-5.826 (-0.514)			
Few Shareholders $_{it-1}$ x Δ Op. Profit $_{it}$			0.010*** (5.010)		
Few Shareholders $_{it-1}$			-3.061 (-0.306)		
Low Concentration $_{it-1}$ x Δ Op. Profit $_{it}$				0.002*** (6.110)	
Low Concentration $_{it-1}$				22.349 (1.205)	
Second Largest Stake $_{it-1}$ x Δ Op. Profit $_{it}$					0.011*** (5.346)
Second Largest Stake $_{it-1}$					13.747 (0.314)
Sales Growth $_{it-1}$	0.001 (0.051)	-0.009 (-0.591)	2.180 (1.310)	6.826*** (2.644)	5.343*** (2.710)
Assets $_{it}$	0.000 (1.276)	0.000 (1.026)	0.000*** (15.429)	0.000*** (6.609)	0.000*** (7.292)
Constant	79.003 (1.207)	42.207 (0.755)	62.985 (0.270)	-58.620 (-0.220)	-13.368 (-0.065)
Firm Controls $_{it}$	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects $_t$	Yes	Yes	Yes	Yes	Yes
Observations	24,069	21,589	13,294	3,465	7,572
Adjusted R-squared	0.01	0.01	0.04	0.08	0.05